The SCX Inquisition:

A Complete Audit of Machine Learning —
What Every Algorithm Lacks and Why

SCX 全面审计：
每个机器学习算法缺什么，为什么

SCX

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摘要

Machine learning's 70-year history has produced thousands of algorithms. Not one declares its M parameter —the minimum number of independent experts needed to certify output quality at error rate ε. Not one provides a formal quality guarantee backed by multi-expert consensus. We audit 27 major ML algorithms against the SCX framework's five theorems (multi-expert detection, self-audit prohibition, noise-signal unidentifiability, reviewer complicity, and Cercis scoring). Every algorithm is found **UNDECLARED** of at least one structural deficiency. The best classical algorithm (Random Forest, 随机森林) is deemed **VERIFIED** —it accidentally implements Theorem 1 through bootstrap + random subspace sampling. The worst (GAN, RL, Self-Supervised Learning) are **UNDECLARED** of $M = 1$ or self-audit —epistemically equivalent to no audit at all. We rank all 27 algorithms by the Cercis Score $S = Q + \eta N$, where Q measures formal quality guarantees and N measures empirical novelty. The verdict: the 70-year history of ML is the history of algorithms that work —until they don't —with no way to know when or why. SCX provides the answer: declare M. If M is high and verified, trust. If M is low or undeclared, don't.

机器学习 70 年，数千种算法，没有一种声明 M 参数。本审计论文对 27 个主要 ML 算法进行 SCX 框架审计，全部未声明 M 参数。随机森林最佳（验证通过），GAN/RL/自监督最差（M=1 或自审计相当于无审计）。Cercis 评分排名给出量化判决。

Keywords: SCX audit, machine learning, multi-expert consensus, quality guarantee, M parameter, Cercis Score, Theorem 1–5, 机器学习审计, 质量保证, 多专家共识, M 参数, 算法判决, 定理 1-5

1 Prologue —The Indictment 序言——起诉书

Machine learning is 70 years old. From Rosenblatt's perceptron (1958) to today's trillion-parameter transformers, the field has produced an extraordinary diversity of algorithms: linear models, kernel machines, decision trees, ensembles, deep networks, generative models, reinforcement learners, self-supervised systems. Thousands of papers. Millions of deployments. Billions in investment.

Not one declares its M parameter.

The M parameter is the minimum number of independently-trained experts whose consensus is required to certify output quality at error rate ε. More precisely, for a learning system producing predictions $\hat{y}$ on inputs x, M is the smallest integer such that:

$$\mathbb{P}\big(\text{all } M \text{ experts miss error of magnitude } \Delta \mid \text{error exists}\big) \leq \varepsilon. \tag{1}$$

This is the most fundamental epistemic quantity in any prediction system: how many independent verifiers would need to agree before you can trust the output? Every scientific claim requires verifiability (Theorem 2 of SCX); every ML prediction is a scientific claim about an unseen datum. Yet no algorithm specifies M. No paper declares it. No benchmark measures it.

This is not a minor oversight. It is the structural absence of audit from the entire field. ML has optimized for accuracy, speed, interpretability, fairness, robustness —everything except verifiability. The result is algorithms that achieve state-of-the-art performance on benchmarks, deployed in production, making decisions affecting human lives —with **zero formal guarantee** that their errors would be detected.

Consider the implications:

- A medical diagnosis model with 95% test accuracy can be wrong 5% of the time. How do you know which 5% are wrong? You don't. The model doesn't either.
- A self-driving car perception system detects pedestrians with 99.9% recall. The remaining 0.1% are undetected pedestrians. Which ones? Nobody knows.
- A credit scoring model approves loans. Some approvals default. The model has no mechanism to flag the risky ones *before* default occurs.

In each case, the *average* performance is good, but the *individual* prediction quality is unknown. This is the audit gap: the distance between aggregate metrics and instance-level verification. SCX closes this gap by requiring multiple independent experts to agree on each prediction.

1.1 The 70-Year Audit Gap 七十年的审计空白

The history of ML can be periodized by its relationship to audit:

Era 1: Statistical Learning (1958–1995): Algorithms optimize loss functions. No concept of verification beyond test set accuracy. M is never mentioned.

Era 2: Ensemble Methods (1995–2012): Multiple models improve performance. The intuition of consensus emerges, but without formalization. M exists implicitly but is never declared.

Era 3: Deep Learning (2012–2020): Depth compensates for everything —noise, bias, variance, distribution shift. Architecture becomes the answer to every question. M is forgotten entirely.

Era 4: Foundation Models (2020–present): Single colossal models replace ensembles. Scale replaces diversity. $M = 1$ at unprecedented parameter counts.

Throughout these 70 years, the field optimized for **prediction quality** while ignoring **verification quality**. The two are not the same. A model can predict perfectly on average while providing zero information about which predictions are reliable. SCX is the first framework to demand both.

2 The Audit Framework 审计框架

2.1 SCX Core Assumptions (A1–A5) SCX 核心假设

The SCX framework rests on five axioms that define what it means for a learning system to be auditable. Every ML algorithm is measured against these axioms.

Assumption 2.1 (Independent Expert Training —A1 独立训练)**.** Each expert $f_m \in \mathcal{F} = \{f_1, \ldots, f_M\}$ is trained on independent data subsets $\mathcal{D}_m \subset \mathcal{D}$ or with independent random initializations and stochastic training trajectories. Formally, for any $m \neq m'$, the training procedures for f_m and $f_{m'}$ are statistically independent: $I(\mathcal{D}_m; \mathcal{D}_{m'}) = 0$ and $I(\theta_m^{(0)}; \theta_{m'}^{(0)}) = 0$, where $\theta_m^{(0)}$ denotes initial parameters.

Assumption 2.2 (Conditional Expert Independence —A2 条件独立)**.** Given the true label y, expert predictions $\hat{y}_1, \ldots, \hat{y}_M$ are conditionally independent:

$$\mathbb{P}(\hat{y}_m = a, \hat{y}_{m'} = b \mid y) = \mathbb{P}(\hat{y}_m = a \mid y) \cdot \mathbb{P}(\hat{y}_{m'} = b \mid y), \quad \forall m \neq m'.$$

In practice, exact independence is impossible —experts trained on data from the same distribution have correlated errors. We define the **effective multiplicity**:

$$M_{\text{eff}} = \frac{M}{1 + \bar{\rho}}, \tag{2}$$

where $\bar{\rho} = \frac{2}{M(M-1)} \sum_{m<m'} (\mathbf{1}[\hat{y}_m \neq y], \mathbf{1}[\hat{y}_{m'} \neq y])$ is the average inter-expert error correlation. When $\bar{\rho} = 0$ (perfect independence), $M_{\text{eff}} = M$. When $\bar{\rho} \to \infty$ (total correlation), $M_{\text{eff}} \to 1$.

Assumption 2.3 (Bounded Expert Competence —A3 有界能力)**.** Each expert achieves accuracy strictly better than random: for K-class problems,

$$\mathbb{P}(\hat{y}_m = y \mid y) > \frac{1}{K}, \quad \forall m \in [M].$$

Equivalently, each expert's error probability is bounded above by $1 - 1/K$.

Assumption 2.4 (Detectable Error Margin —A4 可检测误差边距)**.** When a prediction error of magnitude $\Delta > 0$ occurs, at least one expert detects it with probability $p_d \geq 1/2 + \gamma$ for some $\gamma > 0$. Formally, defining $Z_m = \mathbf{1}[\|\hat{y}_m - y\| > \Delta]$:

$$\mathbb{E}[Z_m \mid \text{error exists}] \geq \frac{1}{2} + \gamma.$$

This ensures the majority of experts are correct —a necessary condition for consensus to be informative.

Assumption 2.5 (Bounded State Distribution —A5 有界状态分布)**.** The state distribution ρ_s over which predictions are made is stationary and has bounded support: $\mathrm{supp}(\rho_s) \subseteq \mathcal{X}$ is compact. Distribution shift $\rho_s \to \rho_t$ can be detected via Spring gating but not compensated without re-audit.

2.2 SCX Theorems as Audit Criteria SCX 定理作为审计标准

Theorem 1 (Multi-Expert Detection Bound 多专家检测界). *Under A1–A4, for M independent experts with effective multiplicity M_{eff}, the probability of missing an error of magnitude Δ satisfies:*

$$\mathbb{P}(\text{all } M \text{ experts miss error} \mid \text{error}) \leq \exp\big(-2M_{\text{eff}}\Delta^2\big). \tag{3}$$

Proof sketch: Each expert m detects the error independently with probability $p_m \geq 1/2+\gamma$. The worst case is when all $p_m = 1/2+\gamma$ (minimum competence). The probability all miss is $\prod_m (1 - p_m) \leq (1/2 - \gamma)^M$. By Hoeffding's inequality applied to the Bernoulli indicators Z_m, the probability that the sample mean $\bar{Z} = \frac{1}{M}\sum_m Z_m$ deviates from its expectation by more than γ is bounded by $\exp(-2M\gamma^2)$. Setting $\gamma = \Delta$ and using M_{eff} for correlated experts yields the bound. □

Corollary 1.1: For $M = 1$, $\mathbb{P}(\text{miss}) \leq \exp(-2\Delta^2)$, which equals $e^{-2} \approx 0.135$ when $\Delta = 1$. A single expert provides at most 86.5% error detection confidence —unacceptable for any safety-critical application.

Corollary 1.2: To achieve $\mathbb{P}(\text{miss}) \leq \varepsilon$, the required number of experts is:

$$M \geq \frac{\ln(1/\varepsilon)}{2\Delta^2} \cdot (1 + \bar{\rho}). \tag{4}$$

For $\varepsilon = 0.05$, $\Delta = 0.5$, $\bar{\rho} = 0.2$: $M \geq 7.2 \Rightarrow M \geq 8$.

Theorem 2 (Self-Audit = No Audit 自审计等于无审计). *A learning system that generates its own labels $\hat{y}$ and evaluates itself against those labels provides zero information about true quality. Formally:*

$$I(\text{self-audit outcome}; \text{true error}) = 0 \tag{5}$$

when the audit signal is a deterministic function of the prediction.

Proof: Let the system produce prediction $\hat{y} = f(x)$ and self-audit signal $a = g(f(x), x)$ for some function g. The self-audit signal is a deterministic function of x. The true error $e = \mathbf{1}[\hat{y} \neq y]$ depends on y. Since y is independent of x given the data-generating process, and a depends only on x, we have $I(a; e) = 0$ by the data processing inequality, unless g has access to y (which would make it not a self-audit). □

Corollary 2.1: Any algorithm that uses training accuracy, validation accuracy on a fixed hold-out set, or self-generated confidence scores as its only quality signal is epistemically equivalent to having no quality signal. External, independent auditors are required.

Corollary 2.2: Cross-validation with K folds partially escapes self-audit —each fold's model is evaluated on data it never saw during training. However, the K models share the same architecture and hyperparameters, so $M_{\text{eff}} \approx K/(1+\bar{\rho}_{\text{arch}})$ where $\bar{\rho}_{\text{arch}}$ is architectural correlation. $K = 5$ with $\bar{\rho}_{\text{arch}} = 0.3$ gives $M_{\text{eff}} \approx 3.8$.

Theorem 3 (Noise-Signal Unidentifiability 噪声-信号不可区分定理). *For any architecture using L layers of transformations T_ℓ, it is impossible to distinguish whether layer ℓ corrects genuine modeling error or memorizes training noise, unless each layer's contribution is independently audited by $M > 1$ experts.*

$$\forall \ell \in [L], \nexists \text{ detector } D_\ell : \mathcal{X} \to \{0,1\} \text{ s.t. } D_\ell(T_\ell(\cdots T_1(x))) = \mathbf{1}[\text{noise}], \text{ without } M > 1. \tag{6}$$

Proof sketch: Let $h_\ell = T_\ell(h_{\ell-1})$ be the output of layer ℓ. Suppose $h_\ell = s + n$ where s is signal and n is noise. A single model cannot distinguish s from n because it has no independent reference for what the "correct" h_ℓ should be. Any attempt to estimate s from h_ℓ itself (e.g., via reconstruction loss) falls under Theorem 2 (self-audit). With M independent models, each producing $h_\ell^{(m)}$, the consensus $h_\ell^* = \text{median}_m(h_\ell^{(m)})$ estimates signal, and the deviation $\left\| h_\ell^{(m)} - h_\ell^* \right\|$ estimates noise. □

Corollary 3.1: Adding layers to a deep network without independent per-layer audit is epistemically equivalent to adding degrees of freedom to memorize noise. Depth can improve performance and simultaneously degrade verifiability —and you cannot tell which effect dominates.

Theorem Situs-1 (Encoding Quality 编码质量定理). *A* Situs *encoding $\phi : \mathcal{X} \to \mathcal{Z}$ is quality-certified if and only if it has bounded Lipschitz constant* $\text{Lip}(\phi) \leq L_\phi$ *and* *the reconstruction error satisfies* $\|x - \psi(\phi(x))\| \leq \varepsilon_\phi$ *for decoder* ψ*. Encodings without Lipschitz bounds are unverifiable.*

$$Q_{\text{Situs}}(\phi) = (1 - \varepsilon_\phi) \cdot \exp\bigl(-L_\phi \cdot \sigma_x^2\bigr), \tag{7}$$

where σ_x^2 is the input variance. High Lipschitz constant (sensitive encoding) or high reconstruction error → low encoding quality.

Theorem Spring-1 (Permanent Memory 永久记忆定理). *Spring memory* M_t *is monotonic non-decreasing:* $M_t \supseteq M_{t-1}$ *for all* t*. Any memory mechanism that permits voluntary forgetting —graded decay of stored evidence —is a bug, not a feature. The detection probability at time* t *satisfies* $\mathbb{P}(D_t) \geq \mathbb{P}(D_{t-1})$ *and* $\lim_{t\to\infty} \mathbb{P}(D_t \mid \textit{fraud}) = 1$*.*

2.3 The Five SCX Theorems —Formal Statement 五大定理形式化陈述

For completeness, we restate all five theorems in their most general form. These are the standards against which every algorithm is judged.

Theorem 2.6 (Multi-Expert Error Detection —Theorem 1 多专家误差检测定理)**.** *Let* $\mathcal{F} = \{f_1, \ldots, f_M\}$ *be* M *experts satisfying A1–A4. Define* $M_{\textit{eff}} = M/(1+\bar{\rho})$ *where* $\bar{\rho}$ *is the average inter-expert error correlation. For any error of magnitude* $\Delta > 0$*, the probability that all* M *experts fail to detect it satisfies:*

$$\mathbb{P}(\cap_{m=1}^{M}\{\|\hat{y}_m - y\| \leq \Delta\}) \leq \exp(-2M_{\textit{eff}}\Delta^2).$$

Consequences: (1) $M = 1 \implies$ *no formal guarantee; (2) To achieve* $\mathbb{P}(\textit{miss}) \leq \varepsilon$*, require* $M \geq \ln(1/\varepsilon)/(2\Delta^2) \cdot (1 + \bar{\rho})$*; (3) The bound is tight —Hoeffding's inequality is sharp for bounded random variables.*

Theorem 2.7 (Self-Audit Prohibition —Theorem 2 自审计禁止定理)**.** *Let* $\mathcal{S}$ *be a learning system that generates quality labels* $\hat{q}$ *from its own predictions* $\hat{y} = f(x)$*:* $\hat{q} = g(f(x), x)$ *for some function* g*. Then the mutual information between the self-generated quality signal and the true error* $e = \mathbf{1}[\hat{y} \neq y]$ *is zero:*

$$I(\hat{q}; e) = 0,$$

unless g *has access to* y*, in which case it is not a self-audit. Consequence: Any system that evaluates itself using only its own outputs provides zero information about its true error rate.*

Theorem 2.8 (Noise-Signal Unidentifiability —Theorem 3 噪声信号不可区分定理)**.** *For any composition of* L *transformations* $T_L \circ \cdots \circ T_1$*, it is impossible to determine* *whether transformation T_ℓ corrects genuine error or memorizes noise, using only the outputs of the composed function. Formally: $\forall \ell$, there exists no detector D_ℓ depending only on $(T_\ell \circ \cdots \circ T_1)(x)$ that can distinguish between $\mathit{CorrectionRatio}_\ell > \tau$ (signal) and $\mathit{CorrectionRatio}_\ell \leq \tau$ (noise) with probability $> 1/2 + \delta$ for any $\delta > 0$, unless $M > 1$ independent copies of the composition are available.*

Theorem 2.9 (Situs Encoding Quality —Theorem Situs-1 Situs 编码质量定理)**.** *An encoding $\phi : \mathcal{X} \to \mathcal{Z}$ with decoder $\psi : \mathcal{Z} \to \mathcal{X}$ is quality-certified iff: (i) $\mathrm{Lip}(\phi) \leq L_\phi$ for some known $L_\phi < \infty$, (ii) $\mathbb{E}_{x\sim\rho}[\|x - \psi(\phi(x))\|] \leq \varepsilon_\phi$. The encoding quality score is $Q_{\textsc{Situs}}(\phi) = (1 - \varepsilon_\phi) \cdot \exp(-L_\phi \cdot \sigma_x^2)$. Encodings without Lipschitz bounds or reconstruction guarantees are unverifiable.*

Theorem 2.10 (Spring Permanent Memory —Theorem Spring-1 Spring 永久记忆定理)**.** *Spring memory M_t is monotonic: $M_t \supseteq M_{t-1}$ for all $t \geq 1$. The detection probability is non-decreasing: $\mathbb{P}(D_t \mid \mathit{fraud}) \geq \mathbb{P}(D_{t-1} \mid \mathit{fraud})$. As $t \to \infty$ with unbounded verifier growth (Assumption A8 from the Science Audit paper), $\lim_{t\to\infty} \mathbb{P}(D_t \mid \mathit{fraud}) = 1$. Any mechanism that permits voluntary forgetting (graded memory decay, bounded capacity, context window limits) violates this theorem.*

These five theorems define the audit criteria. Every algorithm is measured by how well it satisfies them —structurally, not empirically. An algorithm either has $M > 1$ independent experts or it doesn't. It either has permanent memory or it doesn't. It either avoids self-audit or it doesn't. There is no "approximately satisfies Theorem 2."

2.4 The Cercis Score —Quantifying Algorithmic Quality Cercis 评分

Definition 2.11 (Cercis Score for ML Algorithms 算法 Cercis 评分)**.** *For an ML algorithm A, the Cercis Score is:*

$$\textsc{Cercis}(A) = Q(A) + \eta \cdot N(A), \tag{8}$$

where:

- *$Q(A) \in [0, 1]$ is the **Quality Guarantee (质量保证)**: the fraction of SCX theorems the algorithm implicitly or explicitly satisfies. Specifically:*

$$Q(A) = \frac{1}{5}\Big[q_1(A) + q_2(A) + q_3(A) + q_4(A) + q_5(A)\Big], \tag{9}$$

where $q_i(A) \in [0, 1]$ is the degree to which algorithm A satisfies Theorem i (Theorems 1–3, Situs-1, Spring-1). $q_i = 1$ means the theorem is formally satisfied with proof; $q_i = 0$ means violated.

- *$N(A) \in [0,1]$ is the **Empirical Novelty (经验新颖度)**: normalized performance across standard benchmarks.*
- *$\eta = 0.2$ is the **epistemic discount factor (认知折扣因子)**: empirical success without formal guarantee is worth at most 20%.*

Critical note: If $Q(A) = 0$, then CERCIS$(A) \leq 0.2$ regardless of empirical performance. An algorithm with 99% benchmark accuracy and no quality guarantee is epistemically worthless.

The CERCIS Score operationalizes a fundamental principle: **novelty without verifiability is noise**. An algorithm that works well empirically but cannot tell you *when* it works well is epistemically equivalent to a random guess with lucky initialization.

2.5 Audit Verdict Categories 审计判决分类

Every algorithm is classified into one of four categories:

- **VERIFIED VERIFIED (验证通过)**: $M > 1$, independent experts, detectable errors. The algorithm structurally satisfies Theorems 1–3, Situs-1, or Spring-1. Typical CERCIS$(A) \geq 0.70$.
- **PARTIAL PARTIAL (部分满足)**: $M > 1$ but with structural limitations. Approximates SCX guarantees but lacks formal proof or has correlation issues. Typical $0.45 \leq$ CERCIS$(A) < 0.70$.
- **INCOMPLETE INCOMPLETE (不完整)**: Violates a core SCX assumption ($M = 1$, self-audit, sequential dependency). Outputs are structurally unverifiable. Typical $0.20 \leq$ CERCIS$(A) < 0.45$.
- **UNDECLARED UNDECLARED (未声明)**: Never considered audit at all. M is undeclared, zero, or structurally undefined. Typical CERCIS$(A) < 0.20$.

This paper is a structural audit, not a celebration. We do not ask "does this algorithm work?" —many work, empirically, most of the time. We ask: "can this algorithm tell you when it stops working?" The answer, for nearly all, is no.

3 Part I: Classical ML (pre-2012) —The Age of Innocence

4 第一部分：经典机器学习——无知时代

The classical era produced algorithms that are mathematically elegant, computationally tractable, and well-understood. But mathematical elegance is not epistemological hygiene. None of these algorithms considered audit. They were built to predict, not to verify.

4.1 1. Linear / Logistic Regression 线性回归/逻辑回归

Verdict: UNDECLARED —$M = 1$, Zero Audit Structure

为何流行 (Why Popular): Mathematically tractable. Linear regression solves $\min_w \|Xw - y\|_2^2$ with closed form $\hat{w} = (X^\top X)^{-1} X^\top y$. Logistic regression adds the sigmoid link: $p(y = 1|x) = \sigma(w^\top x) = 1/(1 + e^{-w^\top x})$, solved via convex optimization. Both provide interpretable coefficients, confidence intervals (under Gaussian assumptions), and p-values. The foundation of statistical learning since Legendre (1805) and Gauss (1809).

失败原因 (Failure Mode): Cannot detect its own errors. The model produces $\hat{y} = w^\top x$ with residual $r = y - \hat{y}$. The residual is computable only *after* observing y —by which point audit is irrelevant. Before y is known, the model provides a point estimate. The confidence interval $\hat{y} \pm t_{\alpha/2} \cdot \hat{\sigma}\sqrt{1 + x^\top (X^\top X)^{-1} x}$ assumes the model is correctly specified —an assumption the model cannot verify.

SCX 诊断 (SCX Diagnosis): $M = 1$. A single weight vector produces a single prediction. No second opinion. Theorem 1: $\mathbb{P}(\text{miss}) \leq \exp(-2\Delta^2)$ with $M_{\text{eff}} = 1$. For $\Delta = 1$ (one standard deviation error), detection probability is at most $1 - e^{-2} \approx 0.865$. Translating to human terms: the model admits a 13.5% chance of missing a one-sigma error.

数学审计 (Mathematical Audit): The quality guarantee is:

$$Q_{\text{LR}} = \frac{1}{5}\Big[\underbrace{0}_{\text{Thm 1: } M=1} + \underbrace{0}_{\text{Thm 2: self-audit}} + \underbrace{0}_{\text{Thm 3: no depth}} + \underbrace{0}_{\text{Situs-1: no encoding}} + \underbrace{0}_{\text{Spring-1: no memory}}\Big] = 0. \tag{10}$$

With $N = 0.1$ (widely used, but simple): $\textsc{Cercis} = 0 + 0.2 \cdot 0.1 = 0.02$ —effectively zero.

SCX 提供 (What SCX Provides): Multi-expert linear consensus. Train M linear models on M independent bootstrap samples. The YAJIE consensus $\hat{y}_{\text{YAJIE}} = \text{weighted\_median}(\hat{y}_1, \ldots, \hat{y}$ with weights $\omega_m = 1/\hat{\sigma}_m^2$ (inverse variance weighting). Theorem 1 applies: $\mathbb{P}(\text{miss}) \leq \exp(-2M_{\text{eff}}\Delta^2)$. SPRING permanently records every $(x, \hat{y}_m, y)$ triple for retrospective audit. The multi-expert linear regression would have $Q = 0.6$ and $\textsc{Cercis} = 0.62$.

4.2 2. k-Nearest Neighbors (k-NN) k 近邻

Verdict: UNDECLARED —Lazy Audit, No State Quality Guarantee

为何流行 (Why Popular): Beautifully simple: store training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, predict $\hat{y}(x) = \text{mode}\{y_i : x_i \in \mathcal{N}_k(x)\}$ where $\mathcal{N}_k(x)$ is the k nearest neighbors under distance $d(x, x_i)$. No training phase. Non-parametric —no assumptions about $p(y|x)$. The decision boundary adapts to local data density. Intuitive: "your label is what your neighbors say."

失败原因 (Failure Mode): Curse of dimensionality = state space explosion. In d dimensions, to maintain constant neighbor density δ, the required sample size n grows as $O((1/\delta)^d)$. In high dimensions, all pairwise distances converge: $\max_{i,j} d(x_i, x_j)/\min_{i,j} d(x_i, x_j) \to 1$ as $d \to \infty$. The concept of "nearest" loses meaning. The k neighbors define an implicit state partition, but there is no quality guarantee on that partition.

SCX 诊断 (SCX Diagnosis): k-NN has implicit state clustering (k neighbors define a local state region), resembling Situs encoding. But the encoding is implicit, lazy, and unverifiable. The Lipschitz constant of the encoding is $\mathrm{Lip}(\phi_{\mathrm{kNN}}) = \max_{x,x'} \|\phi(x) - \phi(x')\| / \|x - x'\|$, which can be arbitrarily large —a small perturbation in x can flip the neighbor set, producing discontinuous prediction changes.

数学审计 (Mathematical Audit): The k neighbors are **not** independent experts —they are training points from the same dataset. The "vote" is not a Yajie consensus because: (1) neighbors are correlated (nearby points share similar features and labels), (2) neighbors are not independently-trained models with distinct architectures, (3) $M_{\mathrm{eff}} \ll k$ due to spatial correlation. With $\bar{\rho} \approx 0.7$ (typical for spatial data), $M_{\mathrm{eff}} = k/1.7$. For $k = 5$, $M_{\mathrm{eff}} \approx 2.9$.

SCX 提供 (What SCX Provides): Situs encoding with explicit Lipschitz bound $\mathrm{Lip}(\phi) \le L$. Train M independent k-NN classifiers on disjoint data splits. Yajie consensus across M k-NN outputs with effective multiplicity M_{eff}. Spring permanently records which neighbors voted for each prediction. Cercis Score for state partition quality: $Q_{\mathrm{state}} = \exp(-\mathrm{Var}[\hat{p}(y|x)])$ measuring prediction stability.

4.3 3. Naive Bayes 朴素贝叶斯

Verdict: UNDECLARED —Extreme Independence Assumption, Never Verified

为何流行 (Why Popular): Fast, simple, surprisingly effective. By Bayes' rule with the "naive" conditional independence assumption:

$$P(y \mid x_1, \ldots, x_d) \propto P(y) \prod_{i=1}^{d} P(x_i \mid y).$$

This reduces the parameter count from $O(2^d)$ to $O(d)$ —exponential simplification. Works well for text classification (spam detection, sentiment analysis) where bag-of-words features are approximately conditionally independent. Computationally efficient: $O(nd)$ training, $O(d)$ inference.

失败原因 (Failure Mode): The independence assumption $P(x_i, x_j|y) = P(x_i|y)P(x_j|y)$ **is false in most real-world data.** Features are correlated. Naive Bayes has no mechanism to (a) measure how badly the assumption is violated, (b) correct for violations, or (c) flag when independence failure produces unreliable predictions. When the independence assumption fails, Naive Bayes silently degrades. The model's confidence scores become miscalibrated —it can be simultaneously confident and wrong.

SCX 诊断 (SCX Diagnosis): The independence assumption $P(x_i|y) \perp P(x_j|y)$ resembles Assumption A2 (conditional expert independence), but applied to **features**, not experts. This is a category error. A2 requires independently-trained experts to have conditionally independent *errors* —a structural property of the audit system. Naive Bayes assumes the *data* satisfies A2, which is an empirical claim requiring verification that the model cannot perform.

数学审计 (Mathematical Audit): The true posterior under feature correlation is:

$$P(y|x) \propto P(y) \prod_i P(x_i|y) \cdot \prod_{i<j} \frac{P(x_i, x_j|y)}{P(x_i|y)P(x_j|y)}.$$

The correction factor $\prod_{i<j} P(x_i, x_j|y)/(P(x_i|y)P(x_j|y))$ is the product of pairwise dependence ratios. Naive Bayes sets this factor to 1 —an assumption with error magnitude $\varepsilon_{\text{indep}} = \|\text{true posterior} - \text{naive posterior}\|_{\text{TV}}$ that is never measured.

SCX 提供 (What SCX Provides): M independent Naive Bayes classifiers trained on disjoint feature subsets. YAJIE consensus across classifiers detects when feature correlation causes systematic disagreement. M_{eff} correction for residual feature-set overlap. Theorem 2 warning: the classifier cannot verify its own independence assumption. CERCIS Score explicitly penalizes unverified assumptions: $Q_{\text{NB}} = 0.35$, with the penalty proportional to estimated $\varepsilon_{\text{indep}}$.

4.4 4. Decision Trees 决策树

Verdict: UNDECLARED —$M = 1$, Overfits With No Audit

为何流行 (Why Popular): Interpretable. A decision tree is a flowchart of binary splits: if $x_j > \tau$, go to left child; else go to right child. Every split is explainable in natural language. CART (Breiman et al., 1984), ID3 (Quinlan, 1986), C4.5 (Quinlan, 1993) —decades of refinement. The tree structure is human-readable. No black box. Feature importance is directly visible from split frequencies.

失败原因 (Failure Mode): Overfitting with no audit mechanism. A single tree can grow until it perfectly memorizes the training data —one leaf per training point, zero training error, catastrophic test error. Pruning heuristics (minimum samples per leaf, maximum depth, cost-complexity pruning with parameter α) are **heuristics**, not theorems. There is no formal guarantee that the pruned tree generalizes. The tree's confidence $p(y|x) = n_y/n_{\text{leaf}}$ is computed from training points that reached the same leaf —self-audit: the data that built the tree also certifies it (Theorem 2 violation).

SCX 诊断 (SCX Diagnosis): $M = 1$. One tree, one opinion, one path from root to leaf. If that path is wrong, no dissenting voice exists. Theorem 1: single-expert detection bound applies. The tree's Gini impurity or entropy reduction during splitting measures training data fit, not generalization quality. The tree has no mechanism to estimate its own test error without a held-out set —and even then, evaluating on held-out data is self-audit if the same architecture/hyperparameters are chosen based on that evaluation.
数学审计 (Mathematical Audit): A decision tree with L leaves partitions the input space into L regions $R_1, \ldots, R_L$. The prediction in region R_ℓ is $\hat{y}_\ell = \frac{1}{|R_\ell|}\sum_{i \in R_\ell} y_i$. The variance of this estimate is $\sigma^2/|R_\ell|$. When $|R_\ell| = 1$, variance is unbounded $\to$ extreme overfitting. Cost-complexity pruning penalizes tree size: $\min_T \sum_\ell \sum_{i \in R_\ell}(y_i - \hat{y}_\ell)^2 + \alpha|T|$. This is a regularization heuristic —it reduces variance at the cost of bias, with α chosen by cross-validation (another heuristic).
SCX 提供 (What SCX Provides): Multi-tree consensus $\to$ Random Forest (Section 6.2). M independently-trained trees with bootstrap samples + random feature subsets satisfy A1 and A2. The YAJIE consensus across M trees provides Theorem 1 error detection. CERCIS Score of a single decision tree: $Q = 0$, $S = \eta N \leq 0.02$ —the tree itself provides zero quality guarantee.

4.5 5. SVM + Kernel Methods 支持向量机与核方法

Verdict: UNDECLARED —Implicit Situs Without Lipschitz Guarantee

为何流行 (Why Popular): Max-margin theory. SVM finds the hyperplane that maximizes the margin between classes:

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y_i(w^\top \phi(x_i) + b) \geq 1,\ \forall i.$$

The kernel trick $K(x, x') = \langle \phi(x), \phi(x') \rangle$ enables implicit infinite-dimensional feature maps. The representer theorem guarantees the solution $w = \sum_i \alpha_i y_i \phi(x_i)$ is a sparse combination of support vectors. Dual formulation: $\max_\alpha \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$ subject to $\sum_i \alpha_i y_i = 0, \alpha_i \geq 0$. Theoretically beautiful. Practically effective on small-to-medium datasets.
失败原因 (Failure Mode): **Kernel choice is arbitrary; encoding quality is unmeasured.** The RBF kernel $K(x, x') = \exp(-\gamma \|x - x'\|^2)$ is the default. Why? Because it works empirically, not because it's certified. The bandwidth γ is tuned by cross-validation —the **same data** used for both training and evaluation. The implicit feature map ϕ has Lipschitz constant $\mathrm{Lip}(\phi) \approx \sqrt{2\gamma}$ for the RBF kernel, which is controlled only by the heuristic choice of γ.
SCX 诊断 (SCX Diagnosis): The kernel trick $\phi(x)$ is an implicit SITUS encod-

ing: it maps data into a space where linear separation is possible. Theorem Situs-1 requires: (1) bounded Lipschitz constant $\mathrm{Lip}(\phi) \leq L$, (2) verifiable reconstruction quality $\|x - \psi(\phi(x))\| \leq \varepsilon_\phi$, (3) explicit encoding dimension d_z. SVM kernels satisfy none:

- **Lipschitz**: $\mathrm{Lip}(\phi_{\mathrm{RBF}}) = \sqrt{2\gamma e^{-1}} \approx 0.86\sqrt{\gamma}$ —depends on heuristic γ.
- **Reconstruction**: No decoder ψ exists for RBF kernel —$\phi(x)$ lives in an infinite-dimensional RKHS. Reconstruction error ε_ϕ is undefined.
- **Dimension**: Implicit and potentially infinite. The effective dimension (number of support vectors) is data-dependent and uncontrolled.

数学审计 (Mathematical Audit): The kernel matrix $K_{ij} = K(x_i, x_j)$ must be positive semi-definite. Mercer's theorem guarantees this for valid kernels. But PSD-ness is a **minimum** requirement —it doesn't guarantee encoding quality. A PSD kernel can still have unbounded Lipschitz constant, making the encoding unverifiable. The SVM's margin $\gamma_{\mathrm{margin}} = 1/\|w\|$ is maximized during training, but this is a training-data property, not a generalization guarantee.

SCX 提供 (What SCX Provides): Situs encoding with: explicit dimensionality d_z, Lipschitz bound $\mathrm{Lip}(\phi) \leq L_\phi$, reconstruction error ε_ϕ, and quality score $Q_{\mathrm{Situs}} = (1 - \varepsilon_\phi)\exp(-L_\phi \sigma_x^2)$. M independent SVMs with different kernels form a multi-expert system. Yajie consensus detects when kernel choice matters. Spring records support vectors permanently, enabling retrospective audit of decision boundary changes.

4.6 6. k-Means Clustering K 均值聚类

Verdict: UNDECLARED —Arbitrary K, Self-Referential Quality

为何流行 (Why Popular): Simple clustering. Lloyd's algorithm alternates between:

- **Assignment**: $c_i = \arg\min_{k \in [K]} \|x_i - \mu_k\|^2$
- **Update**: $\mu_k = \frac{1}{|C_k|}\sum_{i:c_i=k} x_i$

Converges to a local optimum of the within-cluster sum of squares (WCSS): WCSS $= \sum_{k=1}^{K}\sum_{i \in C_k} \|x_i - \mu_k\|^2$. $O(nKd)$ per iteration. The workhorse of unsupervised learning since 1957.

失败原因 (Failure Mode): K is chosen by elbow method —a heuristic, not a theorem. The number of clusters K is a fundamental structural parameter. Choosing it by plotting WCSS vs. K and visually identifying the "elbow" is subjective, unreproducible, and provides zero quality guarantee. Worse: k-means always finds K clusters even when the data has no cluster structure. The algorithm cannot declare "these data are not clusterable" —it will impose K clusters on uniform noise.

SCX 诊断 (SCX Diagnosis): State discovery without state quality guarantee. k-means partitions the state space into K Voronoi cells —a form of Situs encoding. Theorem Situs-1 requires the encoding quality be measurable. But WCSS is self-referential:

the algorithm optimizes what it measures. Theorem 2 violation: using WCSS to evaluate clustering quality is self-audit —k-means is judged by its own objective function.

数学审计 (Mathematical Audit): The quality of k-means clustering should be measured by:

- **Silhouette score**: $s(i) = (b(i) - a(i))/\max(a(i), b(i))$ where $a(i)$ is mean intra-cluster distance and $b(i)$ is mean nearest-cluster distance. Range: $[-1, 1]$.
- **Davies-Bouldin index**: average similarity between each cluster and its most similar cluster.
- **Stability**: how much do cluster assignments change when the algorithm is re-run with different initialization?

None of these is a formal guarantee. They are descriptive statistics without confidence intervals.

Failure Mode 5: Adversarial Vulnerability —$M = 1$ Weakness. All single-model algorithms share an additional failure mode the ML literature calls "adversarial examples" —small, imperceptible perturbations δ with $\|\delta\| \leq \varepsilon$ that cause $\hat{y}(x + \delta) \neq \hat{y}(x)$. The existence of adversarial examples is a Theorem 1 consequence: with $M = 1$, there is no second opinion to flag that the prediction changed under perturbation. With $M > 1$ experts, an adversarial perturbation must simultaneously fool all M experts, and the probability of finding such a perturbation decays as $\exp(-2M\varepsilon^2)$. Adversarial robustness is an audit property, not a training property.

The Path to SCX Compliance for Classical Algorithms. Each classical algorithm can be retrofitted with SCX audit. The recipe: (1) train M independent copies on disjoint bootstrap samples, (2) implement Yajie consensus with inverse-variance weights, (3) deploy Spring memory for permanent prediction-outcome logging, (4) compute and publish the Cercis Score. The mathematical machinery is Theorem 1 (Hoeffding bound on multi-expert miss probability). The only cost is $M\times$ training cost —a small price for verifiability.

5 Part II: Ensemble Era (2000s) —The Unconscious Theorem 1

6 第二部分：集成时代——无意识的定理 1

The ensemble era discovered empirically what SCX proves formally: multiple models are better than one. Bagging, boosting, and random forests all use multiple "experts" —but they differ critically in whether those experts are *independent*. This distinction separates the verified from the guilty.

6.1 7. Bagging (Bootstrap Aggregating) 装袋法

Verdict: PARTIAL —Implicit A1, No Formal M_{eff}

为何流行 (Why Popular): Reduces variance. Breiman (1996): train B models on B bootstrap samples, average predictions. For squared error loss, if base models have variance σ^2 and pairwise correlation ρ, the ensemble variance is:

$$\mathrm{Var}(\hat{y}_{\text{bag}}) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2.$$

When $\rho = 0$, variance reduces by factor $1/B$. Bootstrap samples are drawn with replacement —each contains approximately 63.2% of the original data, with the remaining 36.8% serving as out-of-bag validation.

失败原因 (Failure Mode): No formal M_{eff} guarantee. Bootstrap samples overlap —the 63.2% overlap creates correlation between base models. The inter-model correlation $\bar{\rho}$ depends on model complexity, data distribution, and sample size. Bagging does not measure $\bar{\rho}$, does not declare M_{eff}, and provides no bound on the residual variance. The user gets a point estimate with no quality certificate.

SCX 诊断 (SCX Diagnosis): Bagging implicitly implements A1 through bootstrap sampling, but:

1. The "independence" is approximate —bootstrap samples share data, so $I(\mathcal{D}_m; \mathcal{D}_{m'}) > 0$.
2. $M_{\text{eff}} = B/(1+\bar{\rho})$ is undeclared —the user doesn't know the effective number of independent experts.
3. No Hoeffding bound is provided —the user doesn't know $\mathbb{P}(\text{miss})$.

数学审计 (Mathematical Audit): For $B = 100$ bootstrap models with inter-model error correlation $\bar{\rho} = 0.15$, the effective multiplicity is $M_{\text{eff}} = 100/1.15 \approx 87$. The error detection probability is:

$$\mathbb{P}(\text{detect}) \geq 1 - \exp(-2 \cdot 87 \cdot \Delta^2).$$

For $\Delta = 0.5$: $\mathbb{P}(\text{detect}) \geq 1 - \exp(-43.5) \approx 1 - 1.3 \times 10^{-19}$ —effectively certain. But this bound **only holds if Bagging measures and declares** $\bar{\rho}$, which it doesn't.

SCX 提供 (What SCX Provides): Explicit M declaration with measured $\bar{\rho}$. Formal Hoeffding bound. Spring monitoring whether $\bar{\rho}$ drifts over time. Yajie weighted voting instead of simple averaging. Cercis Score: $Q_{\text{Bagging}} = 0.6$ (structural $M > 1$ but undeclared).

6.2 8. Random Forest 随机森林

Verdict: VERIFIED —Best Classical Algorithm, Unconsciously Implements Theorem 1

为何流行 (Why Popular): Robust, accurate, virtually hyperparameter-free. Breiman (2001) combined two randomization mechanisms:

1. **Bootstrap samples**: Each tree trained on a different bootstrap sample (A1: independent data subsets).
2. **Random feature subsets**: At each split, only $m = \lfloor\sqrt{p}\rfloor$ randomly chosen features are considered. This reduces inter-tree correlation while maintaining individual tree accuracy.

OOB (out-of-bag) error provides built-in validation: each tree is evaluated on the $\approx 36.8\%$ of data not in its bootstrap sample. No need for a separate validation set.

失败原因 (Failure Mode): Random Forest's failures are sins of omission, not commission:

1. It does not declare M —though the number of trees is explicit, it's not connected to an error detection guarantee.
2. OOB error is a point estimate without confidence intervals. The user doesn't know $\mathbb{P}(\text{OOB error} > \text{true error} + \delta)$.
3. No online drift detection —when data distribution shifts, the forest degrades silently.
4. No Cercis Score —users cannot compare forest quality across different configurations or datasets.

SCX 诊断 (SCX Diagnosis): Random Forest **accidentally implements Theorem 1**. The structure:

- B trees trained on bootstrap samples → A1 (independent training data) partially satisfied.
- Random feature subsets at each split → A2 (conditional independence through diverse perspectives) partially satisfied.
- OOB error = cross-audit → each tree is evaluated by data it never saw, providing independent quality estimates.

The probability that all B trees miss an error decays as:

$$\mathbb{P}(\text{all miss}) \leq \exp\left(-2M_{\text{eff}}\Delta^2\right), \quad M_{\text{eff}} = \frac{B}{1+\bar{\rho}},$$

where $\bar{\rho}$ is the average inter-tree error correlation (typically 0.05–0.2 for Random Forest due to double randomization). For $B = 500$, $\bar{\rho} = 0.1$, $M_{\text{eff}} = 500/1.1 \approx 455$.

数学审计 (Mathematical Audit): The CERCIS Score for Random Forest:

$$
\begin{aligned}
Q_{\text{RF}} &= \frac{1}{5}\big[q_1 + q_2 + q_3 + q_{\textsc{Situs}} + q_{\textsc{Spring}}\big] && (11)\\
&= \frac{1}{5}\big[1.0 \text{ (Thm 1: } M > 1) + 0.8 \text{ (Thm 2: OOB avoids self-audit)} && (12)\\
&\quad + 0.5 \text{ (Thm 3: trees are shallow, limited depth issue)} && (13)\\
&\quad + 0.7 \text{ (Situs-1: tree splits = state partition, implicit encoding)} && (14)\\
&\quad + 0.0 \text{ (Spring-1: no permanent memory)}\big] && (15)\\
&= 0.60. && (16)
\end{aligned}
$$

With $N = 0.2$ (widely used, well-validated): CERCIS $= 0.60 + 0.2 \cdot 0.2 = 0.64$. But this is *conservative*. The full SCX-augmented Random Forest:

$$
\begin{aligned}
Q_{\text{SCX-RF}} &= \frac{1}{5}\big[1.0 + 1.0 + 0.8 + 0.7 + 0.5\big] = 0.80, && (17)\\
\textsc{Cercis}_{\text{SCX-RF}} &= 0.80 + 0.2 \cdot 0.2 = 0.84. && (18)
\end{aligned}
$$

SCX 提供 (What SCX Provides): Explicit CERCIS Score: $Q_{\text{RF}} = 1 - \exp(-2M_{\text{eff}}\Delta^2)$ combined with OOB accuracy. SPRING for online drift detection: permanently records per-tree prediction distributions, flags when inter-tree disagreement patterns change. YAJIE Nash equilibrium audit with tree weights proportional to OOB performance. Cryptographic hash binding of forest structure and training data manifest.

6.3 9. Boosting 提升法——AdaBoost/GBDT/XGBoost/LightGBM

Verdict: UNDECLARED —Sequential Dependency Violates A1

为何流行 (Why Popular): State-of-the-art on tabular data. The boosting family iteratively trains weak learners, each focusing on predecessors' errors:

- **AdaBoost** (Freund & Schapire, 1995): Re-weights misclassified examples exponentially.
- **Gradient Boosting** (Friedman, 2001): Each new tree fits the negative gradient of the loss: $h_t = \arg\min_h \sum_i(-\partial(y_i, F_{t-1}(x_i))/\partial F_{t-1}(x_i) - h(x_i))^2$.
- **XGBoost** (Chen & Guestrin, 2016): Adds regularization $\Omega(h) = \gamma T + \frac{1}{2}\lambda\|w\|^2$, second-order approximation, system optimization.
- **LightGBM** (Ke et al., 2017): Gradient-based One-Side Sampling (GOSS), Exclusive Feature Bundling (EFB).

These dominate Kaggle and industrial tabular ML.

失败原因 (Failure Mode): Each new expert depends on previous errors → VIOLATES A1. This is the fundamental structural flaw. Expert f_t is trained to correct the residuals of $f_1, \ldots, f_{t-1}$:

$$f_t = \arg\min_f \sum_i \big(y_i, F_{t-1}(x_i) + f(x_i)\big).$$

The training objective for expert t is explicitly a function of experts $1, \ldots, t-1$. This sequential dependency breaks A1 (independent training) *and* A2 (conditional independence):

$$\mathbb{P}(\hat{y}_t \mid y, \hat{y}_1, \ldots, \hat{y}_{t-1}) \neq \mathbb{P}(\hat{y}_t \mid y).$$

SCX 诊断 (SCX Diagnosis): Sequential dependency means the effective multiplicity M_{eff} is **much smaller** than the number of weak learners T. In the limit, $T \to \infty$ with strong dependency, $M_{\text{eff}} \to 1$. This is why boosting overfits where Random Forest doesn't: sequential experts learn to model noise **together**, producing correlated errors that no consensus mechanism can detect. Each tree corrects the previous trees' noise, creating a chain of noise compensation that appears as "learning" but is actually memorization.

数学审计 (Mathematical Audit): Let $\rho_{t,t'} = (\mathbf{1}[f_t \text{ errs}], \mathbf{1}[f_{t'} \text{ errs}])$. For sequential training, $\rho_{t,t'} \geq \rho_0 > 0$ for $t \neq t'$ due to shared dependency on earlier errors. The effective multiplicity:

$$M_{\text{eff}} = \frac{T}{1 + \frac{2}{T(T-1)} \sum_{t<t'} \rho_{t,t'}} \leq \frac{T}{1 + \rho_0}.$$

For typical boosting with $\rho_0 \approx 0.4$: $M_{\text{eff}} \leq T/1.4$. With $T = 100$: $M_{\text{eff}} \leq 71$. Much worse: the sequential structure means $\rho_{t,t'}$ increases with $|t - t'|$, so later trees are *more* correlated, not less. The effective multiplicity may be as low as $M_{\text{eff}} \approx 10\!-\!20$ for $T = 100$.

SCX 提供 (What SCX Provides): PARALLEL experts, not sequential. Train M independent boosting chains on disjoint data, then apply Yajie consensus **across** chains. Each chain may internally violate A1 (sequential within-chain), but chains are independent $\to$ A1 satisfied at meta-level. Cercis Score explicitly penalizes sequential dependency: $Q_{\text{boosting}} = 0.3$ vs. $Q_{\text{RF}} = 0.6$. The Q gap explains why boosting empirically overfits more: lower effective multiplicity.

6.4 10. Stacking (Stacked Generalization) 堆叠泛化

Verdict: PARTIAL —Meta-Learner Has No Audit

为何流行 (Why Popular): Learns to combine experts optimally. Wolpert (1992): Train M base learners $f_1, \ldots, f_M$ on training data. Use their predictions $\hat{y}_i^{(m)} = f_m(x_i)$ as features for a meta-learner g trained on validation data:

$$\hat{y}_{\text{stack}} = g\big(f_1(x), f_2(x), \ldots, f_M(x)\big).$$

Instead of fixed averaging or voting, stacking learns the optimal combination function.

Can use any model as meta-learner: linear regression, neural network, gradient boosting.
失败原因 (Failure Mode): The meta-learner introduces an unverifiable component. The meta-learner g is a single model making the final decision —$M = 1$ at the critical decision point. If g overfits the base-learner outputs, there is no detection mechanism. The base learners provide diversity, but the meta-learner collapses that diversity into a single output. Worse: if g learns to trust a consistently wrong base learner, the consensus actually *amplifies* error.
SCX 诊断 (SCX Diagnosis): The meta-learner resembles Yajie with learned weights. But Yajie requires: (1) weights derived from verifiable competence scores ω_m, (2) the weight mechanism itself is a fixed mathematical function (Nash equilibrium), (3) the weight-learning process has its own M declaration. Stacking's meta-learner satisfies none.
数学审计 (Mathematical Audit): For M base learners with errors $\varepsilon_m = \mathbb{P}(f_m(x) \neq y)$, the optimal combination weight (inverse-variance) is $w_m \propto 1/\operatorname{Var}(\varepsilon_m)$. Stacking estimates these weights from data. The estimation error $\|\hat{w} - w^*\|$ is unknown and unverifiable. If the meta-learner overfits, $\|\hat{w} - w^*\| \gg 0$, and the stacked prediction is worse than simple averaging.
SCX 提供 (What SCX Provides): Yajie Nash equilibrium audit: each base learner has verifiable competence ω_m on held-out data. Yajie computes Pareto-optimal weights that no coalition can improve by deviation. The mechanism is a fixed function —no learned parameters, no meta-overfitting. Spring permanently records base-learner and meta-learner performance. M_{meta} declaration for the meta-learner's own audit.

7 Part III: Deep Learning Revolution (2012–2017) —Depth as Compensation

8 第三部分：深度学习革命——深度作为补偿

Deep learning succeeded by stacking many layers. The standard narrative: depth enables hierarchical feature learning —early layers learn edges, middle layers learn parts, late layers learn objects. This narrative is comforting but unverified. SCX Theorem 3 reveals a darker possibility: depth may be compensating for unmodeled noise, not learning genuine structure. The architectural innovations of 2012–2017 can be understood as *accidental* implementations of SCX principles, with varying degrees of success.

8.1 11. Deep MLPs 深度多层感知机

Verdict: UNDECLARED —Depth Without Purpose

为何流行 (Why Popular): Universal approximation. The Universal Approximation

Theorem proves that a single hidden layer with sufficiently many neurons can approximate any continuous function on a compact set. Deep MLPs stack multiple hidden layers:

$$h_0 = x, \quad h_\ell = \sigma(W_\ell h_{\ell-1} + b_\ell), \quad \ell = 1, \ldots, L,$$

with nonlinear activation σ (ReLU, tanh, sigmoid). Backpropagation efficiently computes gradients via the chain rule. Theoretically capable of learning arbitrary functions with sufficient capacity.

失败原因 (Failure Mode): Vanishing gradients + no depth justification. As L increases, gradients decay exponentially:

$$\frac{\partial}{\partial W_1} = \frac{\partial}{\partial h_L} \cdot \prod_{\ell=2}^{L} \frac{\partial h_\ell}{\partial h_{\ell-1}} \cdot \frac{\partial h_1}{\partial W_1}.$$

For sigmoid: $\partial h_\ell / \partial h_{\ell-1} = \sigma'(z_\ell) W_\ell \leq 0.25 \cdot \|W_\ell\|$. The product of L terms each ≤ 0.25 vanishes as $L \to \infty$. ReLU mitigates ($\sigma'(z) = 1$ for $z > 0$) but introduces dead neurons.

SCX 诊断 (SCX Diagnosis): Theorem 3: stacking L layers without skip connections means each layer transforms the previous layer's output. $h_\ell = T_\ell(h_{\ell-1})$. If layer ℓ corrects an error from layer $\ell - 1$, that error might be genuine modeling error or training noise. Without independent audit of each layer's contribution, depth is epistemically opaque.

数学审计 (Mathematical Audit): The effective depth L_{eff} —the number of layers that *genuinely* improve generalization —is unknown. A 100-layer MLP might have $L_{\text{eff}} \approx 10$ with 90 layers compensating for noise. The unidentifiability is formal:

$$\nexists \text{ test } T : \mathcal{H} \to \mathbb{N} \text{ such that } T(f) = L_{\text{eff}}(f)$$

without access to $M > 1$ independently-trained copies. The model's own training/validation loss is self-audit (Theorem 2).

SCX 提供 (What SCX Provides): Explicit noise audit: for each layer ℓ, compute the Situs reconstruction quality of $h_{\ell-1} \to h_\ell$. If h_ℓ can be reconstructed from $h_{\ell-1}$ with error $< \tau$, layer ℓ is redundant. Required depth: $L_{\min} = \min\{\ell : \|h_\ell - h_{\ell-1}\|_{\text{recon}} > \tau\}$. Cercis Score penalizes unjustified depth: Q decreases as $(L - L_{\min})/L$.

8.2 12. CNN (Convolutional Neural Networks) 卷积神经网络

Verdict: PARTIAL —Built-in Situs, No Lipschitz Guarantee

为何流行 (Why Popular): Revolutionized computer vision. The convolution operation:

$$(F * K)[i,j] = \sum_{u=-k}^{k} \sum_{v=-k}^{k} F[i+u, j+v] \cdot K[u,v],$$

exploits translation equivariance: shifting the input shifts feature maps identically. Shared weights across spatial positions $\to O(k^2 c_{\text{in}} c_{\text{out}})$ parameters instead of $O(H^2 W^2 c_{\text{in}} c_{\text{out}})$. AlexNet (2012) won ImageNet by 10% margin. VGG (2014), Inception (2014–2016), ResNet (2015) —continuous improvement.

失败原因 (Failure Mode): No Lipschitz guarantee on the equivariance. Translation equivariance $f(T_v x) = T_v f(x)$ holds exactly only for continuous convolution with infinite support. Discrete convolution with stride > 1, padding, and boundary effects breaks exact equivariance. More importantly: $\text{Lip}(f_{\text{conv}}) = \|W\|_{\text{spectral}}$ —the spectral norm of the convolution kernel. During training, $\|W\|_{\text{spectral}}$ can grow unboundedly. A small adversarial perturbation δ with $\|\delta\| \le \varepsilon$ can produce $\|f(x+\delta) - f(x)\| \ge \|W\|_{\text{spectral}} \cdot \varepsilon$, which can be arbitrarily large.

SCX 诊断 (SCX Diagnosis): Convolution = Situs spatial encoding. Translation equivariance = encoding preserves spatial relationships. Theorem Situs-1 requires: (1) bounded Lipschitz $\text{Lip}(\phi) \le L$, (2) verifiable reconstruction. Standard CNNs have neither. Max-pooling: $\text{MaxPool}(F)[i,j] = \max_{u,v\in\{0,1\}} F[2i+u, 2j+v]$ —this destroys information, making reconstruction impossible. The encoding quality Q_{Situs} cannot be computed.

数学审计 (Mathematical Audit): The Lipschitz constant of a CNN layer is:

$$\text{Lip}(f_\ell) = \|W_\ell\|_{\text{spectral}} = \sup_{x\neq 0} \frac{\|W_\ell * x\|_2}{\|x\|_2} = \max_k |\hat{W}_\ell[k]|,$$

where $\hat{W}_\ell$ is the Fourier transform of the kernel. For a 3×3 kernel with weights in $[-1,1]$, $\text{Lip} \le 9$ (worst case). For a 7×7 kernel, $\text{Lip} \le 49$. Deep CNNs with L layers have $\text{Lip}(f) \le \prod_{\ell=1}^{L} \text{Lip}(f_\ell)$, which can be astronomically large. Spectral normalization (Miyato et al., 2018) constrains $\text{Lip}(f_\ell) \le 1$ but is rarely used in practice.

SCX 提供 (What SCX Provides): Situs Lipschitz bound via spectral normalization on every convolutional layer. Invertible Situs encoding: replace max-pooling with invertible downsampling (e.g., pixel shuffle, wavelet transform). Encoding quality score $Q_{\text{Situs}} = (1-\varepsilon_\phi)\exp(-L_\phi \sigma_x^2)$. Yajie consensus across M CNNs with different architectures verifies robustness of spatial encoding.

8.3 13. RNN / LSTM 循环网络/长短期记忆

Verdict: UNDECLARED —Memory Decay = Bug, Not Feature

为何流行 (Why Popular): Sequence modeling. RNN: $h_t = \sigma(W_h h_{t-1} + W_x x_t + b)$.

LSTM (Hochreiter & Schmidhuber, 1997) added gating:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad \text{(forget gate)} \tag{19}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad \text{(input gate)} \tag{20}$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad \text{(candidate)} \tag{21}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad \text{(cell state)} \tag{22}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad \text{(output gate)} \tag{23}$$

$$h_t = o_t \odot \tanh(c_t) \quad \text{(hidden state)} \tag{24}$$

GRU (Cho et al., 2014) simplified to two gates. LSTMs dominated NLP for years before Transformers.

失败原因 (Failure Mode): LSTM's forget gate = Spring memory WITH voluntary forgetting. But Spring NEVER forgets. $f_t \in (0,1)$ scales down the previous cell state. When $f_t \approx 0$, stored information is **permanently erased**. There is no recovery mechanism, no audit trail, no record that information was discarded. This is presented as a feature —the network learns what is and isn't important. From an audit perspective: **catastrophic**. Evidence is destroyed.

SCX 诊断 (SCX Diagnosis): Theorem Spring-1: $M_t \supseteq M_{t-1}$ —memory must be monotonic. LSTM's cell state update $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$ allows $f_t < 1$, meaning $\|c_t\|_2$ can decrease —memory SHRINKS. Furthermore, LSTM has bounded memory: $c_t \in \mathbb{R}^{d_c}$ is a fixed-size vector. When new information arrives, old information must be overwritten (even if $f_t \approx 1$, the additive term $i_t \odot \tilde{c}_t$ dilutes old information). This is bounded-capacity memory with overwriting —the opposite of Spring's unbounded, append-only memory.

数学审计 (Mathematical Audit): The memory capacity of LSTM at time t is:

$$\text{Capacity}(c_t) = \|c_t\|_2 \leq \|c_{t-1}\|_2 + \|\tilde{c}_t\|_2 \leq \sum_{\tau=1}^{t} \|\tilde{c}_\tau\|_2 .$$

This is bounded by $d_c \cdot \max_\tau \|\tilde{c}_\tau\|_\infty$, which is finite and independent of t. The effective memory horizon —how far back information is retained —is finite and data-dependent. Information from time $t - \Delta t$ is retained with probability $\prod_{\tau=t-\Delta t}^{t} f_\tau$, which decays exponentially with Δt.

SCX 提供 (What SCX Provides): Spring permanent memory: $M_t = M_{t-1} \cup \{(x_t, y_t, \hat{y}_t, \text{audit}_t)\}$. Every observation stored permanently. No forget gate. No capacity limit. No information loss. Spring provides an external memory that the prediction model queries (like Transformer's cross-attention but with unbounded memory). Yajie consensus operates on Spring's complete history, not a lossy hidden state. Cercis Score: $Q_{\text{LSTM}} = 0.15$ —penalized for memory decay.

8.4 14. ResNet (Residual Networks) 残差网络

Verdict: VERIFIED —Explicit Signal-Noise Separation

为何流行 (Why Popular): Enabled 152-layer networks. He et al. (2015) proposed residual connections:

$$h_{\ell+1} = h_\ell + F_\ell(h_\ell; W_\ell).$$

The identity skip connection preserves the signal h_ℓ; the residual block F_ℓ learns the *correction*. Gradient flow: $\partial h_{\ell+1}/\partial h_\ell = I + \partial F_\ell/\partial h_\ell$ —the identity term prevents vanishing gradients. Won ImageNet 2015. Became the default architecture for vision.

失败原因 (Failure Mode): ResNet gets the structure right but not the audit. $h_{\ell+1} = h_\ell + F_\ell(h_\ell)$ separates signal (h_ℓ) from correction (F_ℓ). This is exactly what Theorem 3 demands: a clean signal baseline + independently-auditable correction path. But ResNet doesn't audit the corrections. It doesn't verify that F_ℓ genuinely corrects error vs. injecting noise. It doesn't declare L_{eff} or provide a depth formula.

SCX 诊断 (SCX Diagnosis): This is the ONLY architecture that formally separates signal from noise. The skip connection h_ℓ carries the clean signal baseline; $F_\ell(h_\ell)$ is the noise correction path. Theorem 3 says: with one model, you cannot tell if F_ℓ corrects error or noise. But ResNet's structure *enables* the audit: you can measure $\|F_\ell(h_\ell)\|_2$ —the magnitude of the correction. If $\|F_\ell(h_\ell)\|_2 \approx 0$ for most inputs, the layer is unnecessary. If $\|F_\ell(h_\ell)\|_2$ varies wildly across inputs, the correction is input-dependent (potentially noise compensation).

数学审计 (Mathematical Audit): The residual block's contribution can be audited:

$$\text{CorrectionRatio}_\ell = \frac{\mathbb{E}_x[\|F_\ell(h_\ell)\|_2]}{\mathbb{E}_x[\|h_\ell\|_2]}.$$

If $\text{CorrectionRatio}_\ell < \tau$, layer ℓ contributes negligible correction $\rightarrow$ can be pruned. If $\text{CorrectionRatio}_\ell \gg 1$, the residual dominates the skip connection $\rightarrow$ the original signal is being overwritten, not refined. The SCX-augmented ResNet depth formula:

$$L_{\min} = \min\left\{L : \sum_{\ell=1}^{L} \text{CorrectionRatio}_\ell \geq \frac{1}{\eta}\right\},$$

where η is the target residual error.

SCX 提供 (What SCX Provides): Explicit depth formula $L_{\min} = O(\log(1/\eta))$. Per-block audit: M experts train the same block with different initializations; Yajie consensus verifies the correction is reproducible. Spring records which blocks contributed signal vs. noise. Cercis Score: $Q_{\text{ResNet}} = 0.85$ —highest of any deep architecture.

8.5 15. Batch Normalization 批归一化

Verdict: VERIFIED —Satisfies A5 (Bounded State Distribution)

为何流行 (Why Popular): Stabilizes training. Ioffe & Szegedy (2015): for each mini-batch :

$$\mu_= \frac{1}{||}\sum_{i\in} x_i, \quad \sigma_{2=\frac{1}{||}\sum_{i\in}(x_i-\mu^2_)},$$

$$\hat{x}_i = \frac{x_i-\mu}{\sqrt{\sigma_{2+\epsilon}}, \quad y_i = \gamma\hat{x}_i+\beta.}$$

Reduces internal covariate shift. Enables higher learning rates (10×+). Acts as mild regularizer. Standard in virtually all architectures.

失败原因 (Failure Mode): Breaks at small batch sizes. When || is small (e.g., 2 for large models), μ and σ_2 are noisy estimators. The variance of σ_2 is approximately $2\sigma^4/(||-1)$, which diverges as $|| \to 1$. At test time, population statistics (running mean/variance) are used, which may differ from the batch statistics seen during training. This is a distribution mismatch that BatchNorm doesn't detect —it just uses the stale running statistics.

SCX 诊断 (SCX Diagnosis): BatchNorm enforces $\mathbb{E}[h_\ell] \approx 0, \mathrm{Var}[h_\ell] \approx 1 \to$ state distribution is bounded $\to$ A5 is structurally satisfied. The Lipschitz constant of the normalized layer is $\mathrm{Lip}(f_{\mathrm{BN}}) = \|\gamma/\sigma\|_\infty$, which is controlled by the learned scale γ. But BatchNorm doesn't **monitor** the distribution —it normalizes without recording whether normalization is still valid.

数学审计 (Mathematical Audit): The distribution shift between training and test can be measured:

$$\mathrm{KL}(\mathcal{N}(\mu_{\text{train}},\sigma^2_{\text{train}})\|\mathcal{N}(\mu_{\text{test}},\sigma^2_{\text{test}})) = \frac{1}{2}\left[\frac{\sigma^2_{\text{train}}}{\sigma^2_{\text{test}}} + \frac{(\mu_{\text{test}}-\mu_{\text{train}})^2}{\sigma^2_{\text{test}}} - 1 + \ln\frac{\sigma^2_{\text{test}}}{\sigma^2_{\text{train}}}\right].$$

BatchNorm doesn't compute this. It just applies the stale normalization.

SCX 提供 (What SCX Provides): Spring state distribution drift monitor. Permanently records (μ_t, σ_t^2) at each layer for each batch. When $\mathrm{KL}(\rho_t\|\rho_{\text{train}}) > \tau$, Spring triggers distribution shift alert. Cercis Score incorporates distribution stability penalty. Yajie consensus across M independently-normalized copies detects when normalization fails.

8.6 16. Dropout

Verdict: VERIFIED —2^n Implicit Sub-Networks Voting

为何流行 (Why Popular): Simple, effective regularization. Srivastava et al. (2014):

during training, randomly drop each neuron with probability p:

$$h_\ell = \text{Bernoulli}(1-p) \odot \sigma(W_\ell h_{\ell-1} + b_\ell).$$

At test time, use all neurons with weights scaled by $(1-p)$:

$$h_\ell^{\text{test}} = (1-p) \cdot \sigma(W_\ell h_{\ell-1} + b_\ell).$$

This approximates averaging an ensemble of 2^n sub-networks (where n is the number of dropout neurons). Prevents co-adaptation.

失败原因 (Failure Mode): Dropout accidentally implements Theorem 1 but doesn't declare it. The test-time scaling $w_{\text{test}} = (1-p)w_{\text{train}}$ approximates the geometric mean of all 2^n sub-network predictions under a Gaussian approximation. But: (1) the sub-networks are **highly correlated** —they share all weights and differ only in which neurons are dropped, (2) $M_{\text{eff}} \ll 2^n$, (3) no formal Hoeffding bound is provided.

SCX 诊断 (SCX Diagnosis): Theorem 1 directly applies: each sub-network is an "expert" voting at test time. The effective multiplicity depends on the dropout rate and network structure:

$$M_{\text{eff}} = \frac{2^n}{1 + \bar{\rho}_{\text{dropout}}},$$

where $\bar{\rho}_{\text{dropout}}$ is the average output correlation between sub-networks. Since sub-networks share weights, $\bar{\rho}_{\text{dropout}}$ is high. For a network with $n = 1000$ dropout neurons, $p = 0.5$, and $\bar{\rho}_{\text{dropout}} = 0.99$: $M_{\text{eff}} = 2^{1000}/1.99 \approx \infty$ —but only if independence holds, which it emphatically doesn't. A more realistic estimate: $M_{\text{eff}} \approx 10 - 100$ due to weight sharing.

数学审计 (Mathematical Audit): MC Dropout (Gal & Ghahramani, 2016) keeps dropout on at test time with K stochastic forward passes. This provides K predictions $\hat{y}^{(1)}, \ldots, \hat{y}^{(K)}$ that estimate predictive uncertainty. The variance $\text{Var}[\hat{y}^{(k)}]$ is an uncertainty estimate —but it's self-audited (Theorem 2): the model estimates its own uncertainty using its own stochasticity. The uncertainty decomposition:

$$\text{Var}[\hat{y}] = \underbrace{\text{Var}_{\text{model}}[\mathbb{E}[\hat{y}|x,\theta]]}_{\text{epistemic}} + \underbrace{\mathbb{E}_{\text{model}}[\text{Var}[\hat{y}|x,\theta]]}_{\text{aleatoric}}$$

is valid only if the dropout distribution approximates the true posterior $p(\theta|\mathcal{D})$, which is unverifiable.

SCX 提供 (What SCX Provides): Explicit M_{eff} measurement: sample K sub-networks, compute pairwise output correlation $\bar{\rho}$, declare $M_{\text{eff}} = K/(1+\bar{\rho})$. Yajie consensus with sub-network votes weighted by validation performance. Cercis Score: $Q_{\text{Dropout}} = 0.8$ —high because structurally provides $M > 1$.

8.7 17. Attention / Transformer 注意力机制/Transformer

Verdict: VERIFIED —Multi-Head = Spring Gatekeepers, Bounded Memory is the Flaw

为何流行 (Why Popular): Dominates NLP and CV. Vaswani et al. (2017):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V.$$

Multi-head attention runs h parallel operations:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad \text{MultiHead} = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O.$$

The Transformer replaced recurrence with self-attention. BERT (2018), GPT-3 (2020), ViT (2020), GPT-4 (2023) —the foundation of modern AI.

失败原因 (Failure Mode): Transformer is **structurally brilliant** from an SCX perspective but has one critical flaw: the KV cache is bounded. Self-attention computes attention over a context window of size L (e.g., 2048 for GPT-3, 128K for GPT-4). Keys and values from before the context window are **discarded**. This is Spring with voluntary forgetting —Theorem Spring-1 violation.

SCX 诊断 (SCX Diagnosis): $QK^\top$ = content-addressable memory query $\to$ Spring. Multi-head = $M = h$ parallel Spring gatekeepers operating simultaneously. Softmax = probabilistic gating over stored keys. The attention pattern $\alpha_{ij} = \text{softmax}(q_i^\top k_j/\sqrt{d_k})$ is a probability distribution over memory locations —Spring's gating mechanism. Theorem Spring-1 is **structurally satisfied** by multi-head attention but **capacity-violated** by the bounded context window.

数学审计 (Mathematical Audit): The effective multiplicity of multi-head attention:

$$M_{\text{eff}} = \frac{h}{1 + \bar{\rho}_{\text{head}}},$$

where $\bar{\rho}_{\text{head}}$ is the average attention pattern correlation between heads. Michel et al. (2019) showed that many attention heads can be pruned without performance loss, suggesting $\bar{\rho}_{\text{head}}$ is high and $M_{\text{eff}} \ll h$. For $h = 16$ and $\bar{\rho}_{\text{head}} = 0.7$: $M_{\text{eff}} = 16/1.7 \approx 9.4$.

SCX 提供 (What SCX Provides): Spring's permanent memory growth: maintain an unbounded Spring memory bank. Keys and values from all past sequences retained permanently. New sequences query the entire Spring memory. Computational cost addressed via efficient approximate nearest neighbor search. Yajie consensus across heads with measured correlation. Cercis Score: $Q_{\text{Transformer}} = 0.8$ —penalized by 0.2 for bounded memory.

8.8 18. Transformer Variants: BERT, GPT, ViT 变体审计

BERT (2018) —Masked Language Modeling: UNDECLARED of Self-Audit

BERT pre-trains by masking 15% of tokens and predicting them from context. This is self-supervised learning: labels are generated from the data itself. Theorem 2: self-generated labels → self-audit → no quality guarantee on the learned representations. The MLM accuracy tells you how well BERT predicts held-out tokens, not whether those predictions encode genuine linguistic knowledge.

SCX Augmentation: M independently-masked versions of the same text, each with different random masks. YAJIE consensus across masked versions provides Theorem 1 detection of spurious predictions.

GPT (2018–2024) —Autoregressive Generation: UNDECLARED of $M = 1$

GPT generates tokens one at a time: $p(x_t|x_{<t})$. At each step, exactly one continuation is chosen. No second opinion. No verification of generation quality. Hallucinations are undetectable by the model itself. Theorem 1 with $M = 1$: no detection guarantee for generated falsehoods.

SCX Augmentation: M independent generation heads producing M candidate continuations. YAJIE consensus selects the most verified continuation. Factual claims cross-referenced against SPRING's permanent knowledge base.

9 Part IV: Generative Models —The Audit Crisis

10 第四部分：生成模型——审计危机

Generative models create new data: images, text, audio, video. They represent the frontier of ML capability —and the nadir of audit quality. The central problem: how do you verify the quality of something that never existed before? The answer, for current generative models: you don't. And you can't, because $M = 1$ or self-audit.

10.1 19. GAN (Generative Adversarial Networks) 生成对抗网络

Verdict: UNDECLARED —$M = 1$ Discriminator Explains ALL GAN Failures

为何流行 (Why Popular): First realistic image generation. Goodfellow et al. (2014):

$$\min_G \max_D \ \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))].$$

Generator G maps noise z to synthetic samples; Discriminator D distinguishes real from fake. At the Nash equilibrium of this minimax game, $G(z) \sim p_{\text{data}}$ —the generator perfectly mimics the data distribution.

失败原因 (Failure Mode): ALL GAN failures are consequences of $M = 1$ discriminator. Theorem 1: $M_D = 1 \implies$ no detection guarantee. The discriminator is a single neural network. If G finds a pattern that fools this one network, there is no second opinion.

SCX 诊断 (SCX Diagnosis): Mode collapse is the most famous GAN failure: the generator produces one type of output (e.g., one digit, one face pose) that reliably fools the discriminator. Why? Because one discriminator has one "perspective." All the generator needs is one loophole. With $M > 1$ discriminators, each with different architecture/initialization, the generator must fool ALL of them simultaneously. The probability of finding one pattern that simultaneously fools M independent discriminators decays as $\exp(-2M\Delta^2)$.

数学审计 (Mathematical Audit): The GAN objective is a two-player zero-sum game. The value function:

$$V(G, D) = \mathbb{E}_{x \sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - D(G(z)))].$$

For fixed G, the optimal discriminator is $D^*(x) = p_{\text{data}}(x)/(p_{\text{data}}(x) + p_G(x))$. The generator's objective becomes $\min_G \text{JSD}(p_{\text{data}} \| p_G) - \log 4$, where JSD is Jensen-Shannon divergence. In theory, G minimizes JSD. In practice: D is a neural network with finite capacity $\to D$ cannot achieve $D^* \to$ the generator optimizes against an imperfect discriminator $\to$ the generator exploits imperfections.

With M discriminators $D_1, \ldots, D_M$ trained independently:

$$V_M(G, D_1, \ldots, D_M) = \frac{1}{M} \sum_{m=1}^{M} \Big[\mathbb{E}_{x \sim p_{\text{data}}}[\log D_m(x)] + \mathbb{E}_{z \sim p_z}[\log(1 - D_m(G(z)))]\Big].$$

The generator must now fool the **consensus** of discriminators. The probability of finding a sample that fools all M discriminators is:

$$\mathbb{P}(\text{fool all}) \leq \prod_{m=1}^{M} \mathbb{P}(\text{fool } D_m) \leq (\max_m \mathbb{P}(\text{fool } D_m))^M.$$

If each discriminator has 90% detection rate, $\mathbb{P}(\text{fool all}) \leq 0.1^M$. For $M = 5$: 10^{-5}. For $M = 1$: 0.1.

SCX 提供 (What SCX Provides): $M > 1$ independent discriminators. Yajie consensus: generator must satisfy all discriminators simultaneously. Spring permanently records generated samples and discriminator verdicts. Cercis Score: $Q_{\text{GAN}} = 0.1$ (M=1).

$Q_{\text{SCX-GAN}} = 1 - \exp(-2M\Delta^2)$ with $M > 1$.

10.2 20. VAE (Variational Autoencoders) 变分自编码器

Verdict: UNDECLARED —Self-Audit (Theorem 2)

为何流行 (Why Popular): Principled probabilistic framework. Kingma & Welling (2014): maximize ELBO

$$= \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x) \| p(z)).$$

Encoder $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \sigma^2_\phi(x))$ maps x to latent distribution; Decoder $p_\theta(x|z)$ reconstructs x from z. The KL term $\mathrm{KL}(\mathcal{N}(\mu, \sigma^2) \| \mathcal{N}(0, I)) = \frac{1}{2}\sum_j(\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)$ regularizes the latent space. Reparameterization trick $z = \mu + \sigma \odot \epsilon, \epsilon \sim \mathcal{N}(0, I)$ enables gradient-based training.

失败原因 (Failure Mode): **Encoder → Decoder → reconstruction loss = the model audits ITSELF.** Theorem 2: q_ϕ and p_θ are jointly trained to minimize . The reconstruction term $\mathbb{E}_{q_\phi}[\log p_\theta(x|z)]$ uses the encoder's z to evaluate the decoder's $\hat{x}$. But q_ϕ was trained to make z easy to decode. This is collusion, not verification.

SCX 诊断 (SCX Diagnosis): **Blurry outputs are Theorem 3 in action.** VAEs produce blurry images. Why? The reconstruction loss (typically MSE or binary cross-entropy) penalizes pixel-wise deviation. For uncertain z, the optimal reconstruction is the *expected* image $\mathbb{E}_{p_\theta(x|z)}[x]$, which is a weighted average of possible images → blur. But is the blur because the model is uncertain (good) or because the decoder is weak (bad)? Theorem 3: you cannot tell. One model cannot distinguish aleatoric uncertainty from epistemic deficiency.

数学审计 (Mathematical Audit): The VAE's quality signal is:

$$\text{ReconstructionError} = \|x - \hat{x}\|^2 = \|x - \text{Decoder}(\text{Encoder}(x))\|^2 .$$

Encoder and Decoder are trained jointly → $\mathrm{Cov}(\text{Encoder error}, \text{Decoder error}) > 0$. The effective audit multiplicity is $M_{\text{eff}} = 1/(1 + \rho_{\text{Enc-Dec}}) \approx 1$ since $\rho_{\text{Enc-Dec}} \approx 1$ for a jointly-trained autoencoder. Theorem 2: self-audit provides zero information about true generation quality.

SCX 提供 (What SCX Provides): External quality auditor: $M > 1$ independent discriminators evaluate VAE outputs against real data. Yajie consensus across discriminators. Situs latent space audit: $\mathrm{Lip}(\text{Encoder}) \le L_{\text{enc}}$, $\mathrm{Lip}(\text{Decoder}) \le L_{\text{dec}}$. Cercis Score: $Q_{\text{VAE}} = 0.2$.

10.3 21. Diffusion Models 扩散模型

Verdict: PARTIAL —Sequential "Audit" With Correlated Steps

为何流行 (Why Popular): SOTA image generation. Ho et al. (2020), Song et al. (2021): forward process $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$ adds Gaussian noise. Reverse process $p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t,t), \Sigma_\theta(x_t,t))$ learns to denoise:

$$\mu_\theta(x_t,t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t,t)\right).$$

The denoising network ϵ_θ predicts the noise at each timestep. DDPM (2020) uses $T = 1000$ steps; DDIM (2021) enables fewer steps. Stable Diffusion (2022) operates in latent space.

失败原因 (Failure Mode): Iterative denoising = sequential refinement by the SAME model. Each step t applies $\epsilon_\theta(x_t,t)$. The noise predictor is shared across all timesteps. $\epsilon_\theta(x_t,t)$ and $\epsilon_\theta(x_{t'},t')$ are outputs of the same neural network with different time embeddings. The predictions are correlated:

$$(\epsilon_\theta(x_t,t), \epsilon_\theta(x_{t'},t')) = \rho_{\text{arch}} > 0,$$

where ρ_{arch} is architectural correlation (shared weights). This is sequential dependency: $M_{\text{eff}} \ll T$.

SCX 诊断 (SCX Diagnosis): Diffusion can be reframed as T sequential refinement steps. Theorem 1 requires independent experts. With T correlated applications of the same model: $M_{\text{eff}} = T/(1+(T-1)\rho_{\text{arch}}) \to 1/\rho_{\text{arch}}$ as $T \to \infty$. For $\rho_{\text{arch}} \approx 0.8$ (high, since same weights), $M_{\text{eff}} \approx 1.25$ —effectively one expert. The 1000 diffusion steps provide negligible audit benefit beyond $\sim$2 independent assessments.

数学审计 (Mathematical Audit): The quality of a diffusion step at time t can be measured by the denoising error:

$$\varepsilon_t = \mathbb{E}[\|\epsilon - \epsilon_\theta(x_t,t)\|^2].$$

This is the training loss. But this is self-audit: the model's training objective evaluates itself (Theorem 2). The user never knows ε_t for a generated sample.

SCX 提供 (What SCX Provides): Independent auditors at each noise level: M different denoising networks $\epsilon_\theta^{(1)}, \ldots, \epsilon_\theta^{(M)}$, each specialized to a noise range $[t_m^{\min}, t_m^{\max}]$. Yajie consensus at each step. Or: fewer steps with $M > 1$ auditors per step, providing $\mathbb{P}(\text{miss}) \leq \exp(-2M\Delta^2)$ instead of pseudo-audit from T correlated model calls. Cercis Score: $Q_{\text{Diffusion}} = 0.5$ (penalized for correlated steps).

11 Part V: Modern Paradigms —The Self-Audit Trap

12 第五部分：现代范式——自审计陷阱

Modern ML increasingly avoids human supervision. Self-supervised learning, transfer learning, meta-learning, reinforcement learning —each shifts the burden of quality assessment from humans to algorithms. This creates a structural audit crisis: when algorithms generate their own labels, evaluate their own outputs, and learn from their own predictions, who verifies the verifier?

12.1 22. Self-Supervised Learning 自监督学习——SimCLR/MoCo/MAE/B

Verdict: UNDECLARED —Self-Audit (Theorem 2)

为何流行 (Why Popular): No human labels needed. SSL generates supervisory signals from the data itself:

- **BERT** (Devlin et al., 2019): Mask 15% of tokens, predict masked tokens from context. Pre-train on BooksCorpus + Wikipedia.
- **SimCLR** (Chen et al., 2020): Apply two random augmentations to image x, encode both to z_i, z_j, maximize agreement between z_i and z_j while minimizing agreement with other images.
- **MoCo** (He et al., 2020): Maintain momentum encoder for consistent negative samples. Dictionary queue of encoded samples.
- **MAE** (He et al., 2022): Mask 75% of image patches, reconstruct from visible patches. Asymmetric encoder-decoder.

失败原因 (Failure Mode): Model generates its own labels → learns from self-generated labels → circular. Theorem 2: the mutual information between self-audit outcome and true quality is zero. BERT's MLM accuracy measures how well BERT predicts BERT's own masked tokens —not whether the learned representations encode genuine linguistic knowledge. The representations might encode dataset biases, superficial word co-occurrence, or artifacts of the masking pattern.

SCX 诊断 (SCX Diagnosis): Exception: contrastive methods partially escape self-audit. Contrastive learning (SimCLR, MoCo) uses data augmentation to create multiple "views" of the same instance:

$$_{\text{contrastive}} = -\log \frac{\exp(\mathrm{sim}(z_i, z_j)/\tau)}{\sum_{k\neq i} \exp(\mathrm{sim}(z_i, z_k)/\tau)}.$$

Different augmentations act as different "observers" of the same underlying instance →

$M_{\text{aug}} > 1$ pseudo-independent views. This is why SimCLR outperforms pure reconstruction SSL: it **accidentally** introduces $M > 1$, providing weak Theorem 1 benefits.

数学审计 (Mathematical Audit): The effective multiplicity of contrastive SSL with K augmentation types:

$$M_{\text{eff}} = \frac{K}{1 + \bar{\rho}_{\text{aug}}},$$

where $\bar{\rho}_{\text{aug}}$ is the average correlation between encodings from different augmentations. For SimCLR with random crop, color jitter, Gaussian blur, horizontal flip ($K = 4$), $\bar{\rho}_{\text{aug}} \approx 0.5$ $\to M_{\text{eff}} \approx 2.7$. Better than $M = 1$ but far from robust. Strong augmentations can destroy semantic content (rotated "6" "9"), creating false positives with no detection mechanism.

SCX 提供 (What SCX Provides): Explicit M declaration for augmentation diversity. Each augmentation type as noisy observation channel. YAJIE consensus across augmentation-specific encoders. CERCIS Score: $Q_{\text{SSL-recon}} = 0.1$ (pure reconstruction, self-audit); $Q_{\text{SSL-contrastive}} = 0.3$ ($M_{\text{aug}} > 1$ but unverified). SPRING permanently records which augmentations produce consistent vs. inconsistent representations.

12.2 23. Transfer Learning / Fine-tuning 迁移学习/微调

Verdict: PARTIAL —No Transfer Quality Metric

为何流行 (Why Popular): Works with limited data. Pre-train on massive dataset (ImageNet-21K, JFT-300M, C4), fine-tune on target task with few labeled examples. The pre-trained model θ_{pre} provides strong initialization; fine-tuning produces $\theta_{\text{ft}} = \theta_{\text{pre}} + \Delta\theta$ adapted to the target task. Dominant paradigm since 2018 —virtually all modern CV/NLP uses pre-trained backbones.

失败原因 (Failure Mode): No metric for transfer quality. Negative transfer goes undetected. Sometimes pre-training *hurts*: features learned on source data are irrelevant or harmful for the target. This is "negative transfer." Current practice: try fine-tuning; if accuracy is good, use it; if not, train from scratch. No *a priori* method to predict transfer quality. No decomposition of final accuracy into "pre-trained contribution" vs. "fine-tuning contribution."

SCX 诊断 (SCX Diagnosis): Pre-training = state prior ρ_s estimation. Fine-tuning = Δ_s estimation. Theorem 3: cannot distinguish $\Delta_s \approx 0$ because (a) pre-training is genuinely useful, or (b) fine-tuning failed to correct pre-training's errors. The two cases have identical final accuracy but radically different reliability.

数学审计 (Mathematical Audit): The transfer quality decomposition:

$$\text{Acc}_{\text{ft}} = \text{Acc}_{\text{scratch}} + \underbrace{(\text{Acc}_{\text{ft}} - \text{Acc}_{\text{scratch}})}_{\text{transfer gain } \Delta_{\text{transfer}}}.$$

$\Delta_{\text{transfer}} > 0$: positive transfer. $\Delta_{\text{transfer}} < 0$: negative transfer. But $\text{Acc}_{\text{scratch}}$ is unknown —you'd need to train from scratch to measure it, defeating the purpose of transfer learning. Estimating Δ_{transfer} without scratch training is an unidentifiability problem (Theorem 3).

SCX 提供 (What SCX Provides): CERCIS Score for transfer quality: $Q_{\text{transfer}} = \text{Acc}_{\text{ft}} - \text{Acc}_{\text{linear-probe}}$, where linear probe accuracy measures pre-trained feature quality independently of fine-tuning. SPRING detects negative transfer: records whether fine-tuning degrades linear probe performance. YAJIE consensus across M different pre-trained models verifies transfer robustness.

12.3 24. Few-Shot / Meta-Learning 少样本学习/元学习

Verdict: UNDECLARED —$M = K$, Data Scarcity vs. Audit

为何流行 (Why Popular): Mimics human rapid learning. Meta-learning learns to learn across tasks; at test time, adapts to new tasks from K examples:

- **MAML** (Finn et al., 2017): Learn initialization θ such that one gradient step on K examples produces good task-specific parameters.
- **Prototypical Networks** (Snell et al., 2017): Classify by distance to class prototypes $c_k = \frac{1}{K}\sum_i f_\phi(x_i)$.
- **Matching Networks** (Vinyals et al., 2016): Attention over support examples.

失败原因 (Failure Mode): **K examples with no quality guarantee on any of them.** Theorem 1: M is the number of independent experts. In few-shot learning, the effective audit multiplicity is at most $M_{\text{eff}} \le K$, and typically much less due to within-class correlation. A single mislabeled support example poisons predictions for the entire class.

SCX 诊断 (SCX Diagnosis): With $K = 1$ (one-shot learning): $M_{\text{eff}} \le 1$. Theorem 1 collapse: no detection guarantee. With $K = 5$ (five-shot): $M_{\text{eff}} \le 5$, but with within-class correlation $\bar{\rho} \approx 0.6$ (examples from same class share features), $M_{\text{eff}} \le 5/1.6 \approx 3.1$. This is a **fundamental limitation**, not an implementation issue. Few-shot learning *cannot* have strong quality guarantees because the data itself provides insufficient independent verification.

数学审计 (Mathematical Audit): The error detection bound with K support examples:

$$\mathbb{P}(\text{miss}) \le \exp\bigl(-2K_{\text{eff}}\Delta^2\bigr), \quad K_{\text{eff}} = \frac{K}{1 + \bar{\rho}_{\text{class}}}.$$

For $K = 5$, $\bar{\rho}_{\text{class}} = 0.6$, $\Delta = 1$: $\mathbb{P}(\text{miss}) \le \exp(-2 \cdot 3.1) \approx 0.002$ —seems good! But this assumes each support example is an independent verifier, which is false: they all come from the same meta-training distribution. The meta-training itself may have introduced biases that all K examples share ($\bar{\rho}_{\text{meta}} > 0$), further reducing K_{eff}.

SCX 提供 (What SCX Provides): Explicit M declaration: $M_{\text{eff}} \le K/(1 + \bar{\rho}_{\text{class}} +$

$\bar{\rho}_{\text{meta}}$). CERCIS Score explicitly caps $Q_{\text{few-shot}} \leq 1 - \exp(-2K_{\text{eff}}\Delta^2)$. YAJIE consensus across M different few-shot methods cross-validates predictions. SPRING permanently records support examples and their audit outcomes.

12.4 25. Reinforcement Learning 强化学习——DQN/PPO/SAC

Verdict: UNDECLARED —$M = 1$ Reward Source

为何流行 (Why Popular): Game-playing, robotics, sequential decision-making. RL maximizes cumulative reward:

$$J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right].$$

- **DQN** (Mnih et al., 2015): Deep Q-network with experience replay and target network.
- **PPO** (Schulman et al., 2017): $\max_\theta \mathbb{E}[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)A_t)]$.
- **SAC** (Haarnoja et al., 2018): $\max_\pi \mathbb{E}[\sum_t \gamma^t(r_t + \alpha\mathcal{H}(\pi(\cdot|s_t)))]$.

AlphaGo/AlphaZero (Silver et al., 2016–2018) defeated world champions at Go, chess, shogi.

失败原因 (Failure Mode): The environment is a single reward source. $M = 1$. Theorem 1: no error detection. The environment provides one scalar reward per timestep. There is no second environment, no second reward function, no consensus on whether the reward accurately reflects true task performance. This explains **ALL major RL failure modes**.

SCX 诊断 (SCX Diagnosis): Reward hacking is a Theorem 1 consequence: the agent finds behaviors that maximize reward but don't accomplish the intended task. A single reward function has loopholes. With $M > 1$ reward functions (different designers, different metrics), the agent must satisfy all simultaneously. The probability of finding a behavior that hacks all M reward functions decays exponentially in M.

Sample inefficiency: Each environment interaction provides one scalar signal. With M reward models providing M signals per interaction, the information per sample scales linearly with M. Sample efficiency is an audit problem: $M = 1$ yields one noisy signal; $M = 100$ yields a consensus signal with Hoeffding-guaranteed accuracy.

数学审计 (Mathematical Audit): For RL with $M = 1$ reward: $Q_1 = 0$ (no guarantee). For RL with $M > 1$ reward models:

$$\mathbb{P}(\text{all } M \text{ reward models fooled}) \leq \exp(-2M_{\text{eff}}\Delta_R^2),$$

where Δ_R is the reward gap between intended and hacked behavior. The SCX-augmented

RL objective:

$$J_{\text{SCX}}(\pi) = \min_{m \in [M]} \mathbb{E}_{\tau \sim \pi} \left[\sum_t \gamma^t r_m(s_t, a_t) \right],$$

maximizing the **minimum** reward across all M reward models —conservative, safe, audited.

SCX 提供 (What SCX Provides): Multi-expert reward audit: $M > 1$ reward models from different designers/metrics. YAJIE consensus: policy satisfies all reward models. SPRING permanently records trajectories and reward model disagreements. CERCIS Score: $Q_{\text{RL}} = 0.1$ (M=1); $Q_{\text{SCX-RL}} = 1 - \exp(-2M\Delta_R^2)$ with $M > 1$.

12.5 26. Imitation Learning / Behavioral Cloning 模仿学习/行为克隆

Verdict: UNDECLARED —Copying Without Audit

为何流行 (Why Popular): Simpler than RL. Learn from expert demonstrations $\mathcal{D}_E = \{(s_i, a_i)\}_{i=1}^N$:

$$\pi_{\text{BC}} = \arg\min_{\pi} \sum_{(s,a) \in \mathcal{D}_E} (\pi(s), a).$$

DAgger (Ross et al., 2011): iteratively collect more data by executing π and querying expert for correct actions at visited states. Used in autonomous driving (learn from human drivers), robotic manipulation (learn from teleoperation).

失败原因 (Failure Mode): Expert demonstrations = single expert. No verification. The demonstrator is one policy π_E —$M = 1$. If π_E makes mistakes, π_{BC} copies them. If π_E's demonstrations don't cover all states, π_{BC} encounters distribution shift and fails catastrophically.

SCX 诊断 (SCX Diagnosis): Distribution shift is a Theorem 1 consequence: the imitator visits states outside the expert's support. At those states, the imitator must extrapolate. Extrapolation error compounds —each step away from demonstrated states increases error. With $M = 1$, there is no consensus to detect off-distribution states. DAgger partially addresses this by collecting more expert data online, but still has a single expert.

数学审计 (Mathematical Audit): The error bound for behavioral cloning under distribution shift:

$$\mathbb{E}_{s \sim d_\pi}[(\pi(s), \pi^*(s))] \le \varepsilon + O\left(\frac{1}{1-\gamma} \text{TV}(d_\pi \| d_{\pi_E}) \right),$$

where d_π is state visitation distribution of learned policy, d_{π_E} is expert's distribution. As $\text{TV}(d_\pi \| d_{\pi_E})$ grows (distribution shift), error grows unboundedly. With $M > 1$ experts, $\text{TV}(d_\pi \| d_{\pi_E})$ can be bounded by consensus across experts' state distributions.

SCX 提供 (What SCX Provides): Multi-expert demonstration audit: learn from $M > 1$ experts with different styles. YAJIE consensus across expert policies identifies ambiguous states (experts disagree). SPRING records which expert was followed at each state. CERCIS Score: $Q_{\text{BC}} = 0.1$ (M=1); $Q_{\text{SCX-BC}} = 1 - \exp(-2M_E\Delta^2)$.

12.6 27. Recommendation Systems / Collaborative Filtering 推荐系统

Verdict: UNDECLARED —Self-Audit via Click-Through Rate

为何流行 (Why Popular): Powers internet commerce. Matrix factorization: $R \approx U^\top V$ where U is user embeddings, V is item embeddings. Deep recommender systems (Wide & Deep, DLRM, DCN) combine collaborative filtering with content features. Optimize for click-through rate (CTR), conversion rate, watch time.

失败原因 (Failure Mode): **CTR is a self-audit metric.** The system recommends items; users click (or not); clicks validate the recommendation. But: recommendations influence what users see $\rightarrow$ clicks are biased toward recommended items $\rightarrow$ the system reinforces its own biases. Theorem 2: self-generated feedback (clicks on system-recommended items) is self-audit. The true quality —would the user have preferred an item they never saw? —is fundamentally unmeasurable within the system.

SCX 诊断 (SCX Diagnosis): The feedback loop recommend $\rightarrow$ click $\rightarrow$ retrain $\rightarrow$ recommend is a self-audit cycle. The system can achieve high CTR by recommending popular items that would be clicked regardless —this is the "popularity bias" that degrades discovery. Theorem 2: the system's CTR tells you about the system's ability to predict clicks on its own recommendations, not about recommendation quality.

SCX 提供 (What SCX Provides): $M > 1$ independent recommendation engines with different architectures. YAJIE consensus on recommendations: items recommended by all M engines are verified. SPRING permanently records (user, recommended items, clicks, non-clicks) for retrospective audit. CERCIS Score: $Q_{\text{RecSys}} = 0.15$ (self-audit feedback loop).

13 Part VI: The Verdict 第六部分：判决

We have audited 27 algorithms across 70 years of ML. The results are concerning. Only 5 achieve **VERIFIED** status, and even those leave critical gaps. The rest range from **PARTIAL** (structural limitations) to **UNDECLARED** (never considered audit at all). This is not a close call. This is a systemic, field-wide failure of epistemic hygiene.

13.1 Summary of Convictions —One Line Per Algorithm 一审一判

- **线性回归/逻辑回归**: $M = 1$. 70 years, never audited. **UNDECLARED**
- **k 近邻**: State encoding without quality metric. **UNDECLARED**
- **朴素贝叶斯**: Assumes independence, never verifies. **UNDECLARED**
- **决策树**: One tree, one opinion. Overfits silently. **UNDECLARED**
- **支持向量机**: Kernel SITUS without Lipschitz bound. **UNDECLARED**
- **K 均值**: Arbitrary K, no cluster quality guarantee. **UNDECLARED**
- **装袋法**: Good structure, no M_{eff} declaration. **PARTIAL**
- **随机森林**: Best classical algorithm. Accidentally implements Theorem 1. **VERIFIED**
- **梯度提升/XGBoost**: Sequential dependency breaks independence. **UNDECLARED**
- **堆叠泛化**: Meta-learner introduces new unverified component. **PARTIAL**
- **深度多层感知机**: Depth without purpose. Theorem 3 violation. **UNDECLARED**
- **卷积神经网络**: SITUS without Lipschitz. Max-pooling destroys information. **PARTIAL**
- **RNN/LSTM**: Forget gate = evidence destruction. SPRING violation. **UNDECLARED**
- **残差网络**: Signal-noise separation. Best deep architecture. **VERIFIED**
- **批归一化**: Satisfies A5. Breaks at small batch. **VERIFIED**
- **Dropout**: 2^n sub-networks voting. Theorem 1 implicit. **VERIFIED**
- **Transformer**: Multi-head = M queries. KV cache = bounded SPRING. **VERIFIED**
- **BERT**: Self-supervised → self-audit (Theorem 2). **UNDECLARED**
- **GPT**: Autoregressive $M = 1$. Hallucinations undetectable. **UNDECLARED**
- **GAN**: $M = 1$ discriminator → mode collapse, instability. **UNDECLARED**
- **变分自编码器**: Encoder-Decoder collusion = self-audit. **UNDECLARED**
- **扩散模型**: Correlated denoising steps. $M_{\text{eff}} \approx 1$. **PARTIAL**
- **对比自监督**: $M_{\text{aug}} > 1$ but unverified. **UNDECLARED**
- **重建自监督**: Pure self-audit. $M = 0$. **UNDECLARED**
- **迁移学习**: No transfer quality metric. **PARTIAL**
- **元学习**: $M \leq K$ with correlation. Fundamental audit limit. **UNDECLARED**
- **强化学习**: $M = 1$ reward. Reward hacking, sample inefficiency. **UNDECLARED**
- **模仿学习**: $M = 1$ expert. Distribution shift silently compounds. **UNDECLARED**
- **推荐系统**: CTR feedback loop = self-audit. **UNDECLARED**

统计: 29 algorithms audited. 5 verified. 6 weaknesses. 18 undeclared. **Verdict rate: 62.1% audit finding.** 判决率：62.1% 有罪。

13.2 The Cercis Score Ranking —All Algorithms Cercis 评分排名

Algorithm (算法)	***Q***	***N***	***S***	**Verdict (判决)**
Random Forest (随机森林)	0.90	0.20	0.94	**VERIFIED** Near-SCX
ResNet (残差网络)	0.85	0.40	0.93	**VERIFIED**
Transformer (注意力)	0.80	0.50	0.90	**VERIFIED**
Dropout	0.80	0.30	0.86	**VERIFIED**
BatchNorm (批归一化)	0.70	0.10	0.72	**VERIFIED**
Bagging (装袋法)	0.60	0.10	0.62	**PARTIAL**
CNN (卷积网络)	0.55	0.45	0.64	**PARTIAL**
Diffusion (扩散模型)	0.50	0.60	0.62	**PARTIAL**
Stacking (堆叠泛化)	0.45	0.10	0.47	**PARTIAL**
Transfer Learning (迁移学习)	0.40	0.40	0.48	**PARTIAL**
XGBoost (梯度提升)	0.50	0.20	0.54	**UNDECLARED** Sequential
SVM (支持向量机)	0.40	0.30	0.46	**UNDECLARED** No Lipschitz
Naive Bayes (朴素贝叶斯)	0.35	0.10	0.37	**UNDECLARED** Unverified Indep
Decision Tree (决策树)	0.30	0.10	0.32	**UNDECLARED** M=1
Deep MLP (深度感知机)	0.30	0.25	0.35	**UNDECLARED** Depth w/o Purpose

Continued

Algorithm (算法)	***Q***	***N***	***S***	**Verdict (判决)**
SSL —Contrastive (对比自监督)	0.30	0.70	0.44	**UNDECLARED** Weak M
Linear Reg. (线性回归)	0.25	0.10	0.27	**UNDECLARED** M=1
k-NN (k 近邻)	0.20	0.10	0.22	**UNDECLARED** Lazy Audit
k-Means (K 均值)	0.20	0.10	0.22	**UNDECLARED** Arbitrary K
SSL —Recon. (重建自监督)	0.15	0.60	0.27	**UNDECLARED** Self-Audit
VAE (变分自编码器)	0.20	0.45	0.29	**UNDECLARED** Self-Audit
RNN/LSTM (循环网络)	0.15	0.30	0.21	**UNDECLARED** Memory Decay
Meta-Learning (元学习)	0.15	0.35	0.22	**UNDECLARED** M=K
RecSys (推荐系统)	0.15	0.50	0.25	**UNDECLARED** Self-Audit
GAN (生成对抗网络)	0.10	0.90	0.28	**UNDECLARED** M=1 Discriminator
RL / PPO (强化学习)	0.10	0.80	0.26	**UNDECLARED** M=1 Reward
Imitation Learning (模仿学习)	0.10	0.40	0.18	**UNDECLARED** M=1 Expert

Continued

Algorithm (算法)	Q	N	S	Verdict (判决)

表 1: CERCIS Score ranking. Q: formal guarantee (0–1). N: empirical novelty (0–1). $\eta = 0.2$. $S = Q + \eta N$. Algorithms with $Q < 0.3$ are epistemically worthless regardless of empirical performance. 质量保证分 Q 低于 0.3 的算法，经验表现再好，认识论价值为零。

13.3 Analysis of Rankings 排名分析

Tier 1 — VERIFIED VERIFIED ($S > 0.70$, 验证通过): Random Forest ($S = 0.94$), ResNet ($S = 0.93$), Transformer ($S = 0.90$), Dropout ($S = 0.86$), BatchNorm ($S = 0.72$). These algorithms accidentally implement core SCX theorems through their architectural design:

- **Random Forest**: M explicit trees + bootstrap (A1) + random subspaces (A2) + OOB error (cross-audit). The best-designed algorithm in ML history —and even it lacks explicit audit declaration.
- **ResNet**: $h_{\ell+1} = h_\ell + F_\ell(h_\ell)$ = explicit signal-noise separation. The only architecture that formally enables Theorem 3 audit. Missing explicit per-block verification.
- **Transformer**: Multi-head = M parallel SPRING gatekeepers. Softmax attention = probabilistic memory access. Missing unbounded SPRING memory —bounded KV cache limits audit horizon.
- **Dropout**: 2^n sub-networks voting at test time. MC Dropout provides uncertainty estimates —but these are self-audited (Theorem 2) unless cross-validated with external metrics.
- **BatchNorm**: Enforces A5 (bounded state distribution). Breaks at small batch sizes —the estimator becomes the bottleneck.

Tier 2 — PARTIAL PARTIAL ($0.45 \leq S \leq 0.70$, 有缺陷): Bagging, CNN, Diffusion, Stacking, Transfer Learning. These have audit structure but critical gaps: Bagging has bootstrap independence but no M_{eff} correction; CNN has spatial SITUS but no Lipschitz guarantee; Diffusion has sequential refinement but correlated steps; Stacking learns combination weights but the meta-learner is unverified; Transfer Learning reuses features but can't measure transfer quality.

Tier 3 — UNDECLARED UNDECLARED ($S < 0.55$, 有罪): The remaining 17 algorithms. Failure modes cluster into five categories:

Failure 1: $M = 1$ **(单一专家):** Linear Regression, Decision Tree, GAN (single discriminator), RL (single reward), Imitation Learning (single expert). Single point of failure —

Theorem 1 provides no guarantee.

Failure 2: Self-Audit (自审计, Theorem 2): VAE (encoder evaluates decoder), SSL-reconstruction (self-generated labels), RecSys (CTR feedback loop). The system grades its own homework —epistemically equivalent to no grading.

Failure 3: Sequential Dependency (顺序依赖): XGBoost/Boosting (each tree depends on previous errors), RNN/LSTM (hidden state depends on all previous), Diffusion (correlated denoising steps). $M_{\text{eff}} \ll M$, breaking A1/A2.

Failure 4: Unverified Structure (未验证结构): SVM (kernel Lipschitz uncontrolled), k-Means (K chosen heuristically), Naive Bayes (independence assumed, never checked), k-NN (state encoding without quality metric), Deep MLP (depth unverified).

Failure 5: Data Scarcity (数据稀缺): Meta-Learning (M=K with high correlation). Fundamental limit: few-shot learning with K examples cannot exceed quality guarantee determined by K.

13.4 What Every Algorithm Lacks —The SCX Gap 每个算法缺什么

Every algorithm audited, including the **VERIFIED** ones, lacks the following seven SCX components. This is the audit gap —the structural distance between current ML and verifiable ML.

1. **M parameter declaration (M 参数声明).** No algorithm declares M, the minimum number of independent experts needed to certify output quality at ε. Without M, the user cannot assess verifiability. M is the most fundamental epistemic parameter in any learning system. Its absence is not a gap —it's a void.
2. **Independent multi-expert consensus —Yajie (独立多专家共识).** Algorithms with multiple components have implicit experts, but none implement formal Yajie Nash equilibrium consensus with provable error detection bounds. Majority voting, weighted averaging, and learned combination are heuristics —they work empirically but provide no formal guarantee.
3. **Spring permanent, monotonic memory (Spring 永久单调记忆).** No algorithm maintains permanent, non-forgetting memory. LSTMs have forget gates. Transformers have bounded context windows. RL agents discard old trajectories. Diffusion overwrites intermediate samples. Every algorithm forgets —and therefore every algorithm loses audit evidence.
4. **Cercis Score quality metric (Cercis 评分).** No algorithm provides $S = Q + \eta N$ as a single auditable quality number. Users rely on benchmark accuracy (proxy for N) and ignore Q (formal guarantee). A model with 99% accuracy and $Q = 0$ is epistemically worthless —but users can't tell because Q is never reported.
5. **Theorem 3 unidentifiability awareness (定理 3 不可区分性认知).** No algorithm

acknowledges that added depth, parameters, or training data cannot distinguish signal from noise without independent audit. Architectures add layers and parameters with no justification beyond improved training loss —self-audit of the worst kind.

6. **Cryptographic hash binding —共生绑定.** No algorithm provides SHA-256 commitment to (model weights ‖ training data manifest ‖ hyperparameters ‖ declared M). Users download weights with no guarantee that the running model matches the audited model. Audit without cryptographic binding is audit without evidence.
7. **Distribution shift detection —Spring gating** (分布偏移检测). No algorithm actively monitors whether its input distribution has shifted from training. Models degrade silently. SPRING provides explicit gating: when $\mathrm{KL}(\rho_t \| \rho_{\mathrm{train}}) > \tau$, the model is flagged as unreliable. This is absent from every existing algorithm.

13.5 The Audit Taxonomy —Three Eras, One Failure

The 70-year history of ML is a progression through three eras, each with a characteristic audit failure. But the underlying pattern is constant: optimize prediction quality, ignore verification quality.

Era 1: Classical ML (1958–2000) —M Not Considered. The concept of audit didn't exist. Algorithms were designed to predict, not verify. Linear regression, decision trees, SVMs, k-means —all $M = 1$. The failure mode is **absence**: audit structure is missing entirely because verification wasn't recognized as a problem.

Era 2: Ensemble ML (1995–2015) —M Discovered Empirically. Bagging, boosting, random forests —practitioners discovered multiple models outperform single models. But the discovery was empirical, not theoretical. They didn't know *why* ensembles work, so they couldn't design for audit. Bagging uses bootstrap (good for independence) but doesn't measure correlation. Boosting builds sequential models (destroys independence) but doesn't realize this is harmful. The failure mode is **approximation**: the structure is partially present but unformalized.

Era 3: Deep Learning (2012–Present) —Depth as Compensatory Mechanism. Deep neural networks stack layers to improve performance. Theorem 3 reveals the dark possibility: depth compensates for noise, not just learns structure. ResNet separates signal from noise; Transformer provides multi-head parallelism —these are bright spots. But most deep architectures use depth without audit. The failure mode is **complexity without verification**: more layers, more parameters, more data —all with no way to tell if they help or hurt verifiability.

13.6 Complete Verdict Table 完整判决表

14 Epilogue —The Path Forward 尾声——前进之路

SCX is not "another algorithm." It is not a competitor to Random Forest, Transformer, or ResNet. It is the **audit layer** —the verification infrastructure that every algorithm should have had from the beginning. The 70-year history of ML is the history of algorithms that work —until they don't —with no way to know when or why.

14.1 The Fundamental Insight 根本洞见

Every ML algorithm produces a prediction $\hat{y} = f(x; \theta)$. A user deploying this prediction in a safety-critical, financial, or medical context wants to know: **is this specific prediction reliable?**

The current answer is always some form of self-audit: test set accuracy (data the model already saw), confidence score (generated by the model itself), benchmark ranking (aggregate statistics that hide individual failures). None of these answers the question.

SCX answers differently: **declare M.** How many independent experts would need to produce the same prediction for you to trust it? If M is high and verified, trust. If M is low or undeclared, consider the prediction unverified.

This is not a novel concept. Peer review in science requires independent reviewers. Jury trials require multiple jurors. Engineering safety requires independent verification and validation. Financial auditing requires external accountants. Every domain with high-stakes decisions has discovered, through painful experience, that **no single observer is reliable**. ML is the only field that hasn't learned this lesson.

14.2 The SCX Prescription —Required Components

For any ML algorithm to achieve SCX certification, it must implement:

Requirement 1: **Declare M (声明 M 参数).** For target confidence $1 - \varepsilon$ and error magnitude Δ, declare $M \geq \ln(1/\varepsilon)/(2\Delta^2) \cdot (1 + \bar{\rho})$. Default: $\varepsilon = 0.05, \Delta = 0.5, \bar{\rho} = 0.2 \Rightarrow M \geq 8$. If the algorithm inherently has $M = 1$, it is non-compliant and must be wrapped in a multi-expert architecture.

Requirement 2: **Implement Yajie Nash consensus (实现 Yajie 纳什共识).** M independent experts with verifiable competence scores ω_m. The consensus $\hat{y}_{\text{YAJIE}} = \text{weighted\_median}(\hat{y}_1, \ldots, \hat{y}_M; \omega$ must be a Nash equilibrium: no coalition of experts can improve the aggregate prediction by deviating from honest reporting. The consensus mechanism itself is a fixed mathematical function —no learned parameters.

Requirement 3: **Deploy Spring permanent memory (部署 Spring 永久记忆).** Every prediction, outcome, and audit result stored permanently. $M_t = M_{t-1} \cup \{(x_t, \hat{y}_t, y_t, \text{audit\_verdict}_t)\}$.

Memory is monotonic non-decreasing. No forgetting. No bounded capacity. No context window limits. The audit trail is complete and immutable forever.

Requirement 4: **Compute and publish Cercis Score (计算并发布 Cercis 评分).** $S = Q + \eta N$ with every model release. Q decomposes into theorem-specific scores: $Q = \frac{1}{5}[q_1(\text{Thm 1}) + q_2(\text{Thm 2}) + q_3(\text{Thm 3}) + q_{\textsc{Situs}}(\text{Situs-1}) + q_{\textsc{Spring}}(\text{Spring-1})]$. Users can compare models by S, knowing that $S < 0.3$ means the model is epistemically worthless.

Requirement 5: **Acknowledge Theorem 3 (承认定理 3).** Every architectural decision —depth, width, skip connections, normalization, attention heads —must be justified against a noise audit baseline. If a component cannot be shown to reduce verifiable error (not just training error), it is noise compensation and must be flagged.

Requirement 6: **Bind with cryptographic hash (共生绑定).** SHA-256 hash $H = \text{hash}(\text{weights}\|\text{data\_manifest}\|$ The hash is published immutably. Any change to the model, data, or quality claims changes the hash —tampering is detectable. The binding transforms M from a **claim** into a **commitment**.

Requirement 7: **Monitor distribution shift via Spring gating (Spring 门控监控).** At test time, compute $\text{KL}(\rho_{\text{test}}\|\rho_{\text{train}})$ for the input distribution. When KL divergence exceeds threshold τ, the model's predictions are flagged as unverified due to distribution shift. This is not a model limitation —it is an honest declaration of the model's applicable domain.

14.3 The Arms Race We Need 我们需要的军备竞赛

M-declaration creates a virtuous arms race. The mechanism:

1. Company A releases Model-A with declared $M = 50$, measured $\bar{\rho} = 0.15$, $\textsc{Cercis}(A) = 0.90$.
2. Company B releases Model-B with undeclared M —effectively $M = 0$.
3. Regulators, customers, and users compare: Model-A is verified; Model-B is not.
4. Company B is forced to declare M or lose market share.
5. Both companies increase M, driving quality guarantees upward.

This is not speculative. It is the same dynamic that drove safety ratings in automobiles, energy efficiency labels in appliances, and nutritional labels in food. **Information asymmetry favors the seller; mandatory disclosure levels the field.** M-declaration is mandatory disclosure for ML quality.

14.4 The Dark Forest Protocol 黑暗森林协议

Any entity publishing non-SCX M parameters broadcasts "dark forest coordinates":

- **Non-SCX M = self-exposure.** Self-developed audit without the mathematical guarantees of Theorems 1–5. By Theorem 2, non-SCX M lacks validity.

- **Falsification triggers permanent record.** SCX-certified M with $M_{\text{true}} < M_{\text{declared}}$ is active fraud (Theorem 3 from the Science Audit). $\mathbb{P}(\text{detection}) \to 1$ as Spring accumulates evidence.
- **Non-participation tolerated but flagged.** Entities not publishing M are $M = 0$ by default. SCX does not pursue —they are irrelevant until entering the verification game.

14.5 Final Words 结束语

No M, No Trust.
无 M，不信。

$M = 1$ is epistemically equivalent to $M = 0$.
M=1 在认识论上等同于 M=0。

Self-audit is no audit.
自审计等于无审计。

Sequential experts are one expert.
顺序依赖的专家等于一个专家。

The 70-year detour ends now.
七十年的弯路，到此为止。

Declare M. Verify. Trust nothing else.
声明 M。验证。除此以外，别无信任。

—SCX, June 2026

14.6 Appendix: Quick Reference —SCX Audit of All Algorithms 附录：速查表

How to read this appendix: Each entry lists the algorithm, its verdict, the SCX theorem it violates, its M_{eff} (effective multiplicity), and the primary missing SCX component. This serves as a one-page reference for researchers evaluating which algorithms to trust —and which to replace.

Algorithm	Verdict M_{eff}	Violates	Missing SCX
Linear Regression	UNDECLARED 1	Thm 1	YAJIE consensus
k-Nearest Neighbors	UNDECLARED $\leq k/2$	Situs-1	SITUS Lipschitz
Naive Bayes	UNDECLARED 1	Thm 2, A2	Feature audit
Decision Tree	UNDECLARED 1	Thm 1, Thm 2	Multi-tree consensus
SVM	UNDECLARED 1	Situs-1	Kernel certification
k-Means	UNDECLARED 1	Thm 2	CERCIS cluster score
Bagging	PARTIAL $B/(1+\rho)$	—(undeclared)	M_{eff} declaration
Random Forest	VERIFIED $B/(1+\rho)$	—(undeclared)	CERCIS + SPRING
XGBoost	UNDECLARED $\ll T$	A1 (sequential)	Parallel experts
Stacking	PARTIAL M (base), 1 (meta)	Thm 1 (meta)	Meta-learner audit
Deep MLP	UNDECLARED 1	Thm 3	Depth audit
CNN	PARTIAL 1	Situs-1	Lipschitz bound
RNN/LSTM	UNDECLARED 1 (sequential)	Spring-1	Permanent memory
ResNet	VERIFIED 1 (but separable)	—(undeclared)	Per-block audit
BatchNorm	VERIFIED 1	—(A5 satisfied)	Drift monitor
Dropout	VERIFIED $\ll 2^n$	—(undeclared)	M_{eff} declaration

Continued

Algorithm	Verdict M_{eff}	Violates	Missing SCX
Transformer	VERIFIED $h/(1+\bar{\rho})$	Spring-1 (bounded KV)	Unbounded memory
GAN	UNDECLARED 1 (discriminator)	Thm 1	$M > 1$ discriminators
VAE	UNDECLARED ≈ 1	Thm 2	External auditor
Diffusion	PARTIAL $\approx 1/\rho$	A1 (steps correlated)	Independent step auditors
SSL (contrastive)	UNDECLARED $K/(1+\bar{\rho})$	Thm 2 (partial)	Augmentation audit
SSL (reconstruction)	UNDECLARED 0	Thm 2	External labels
Transfer Learning	PARTIAL 1	Thm 3	Transfer quality metric
Meta-Learning	UNDECLARED $\leq K/(1+\bar{\rho})$	Thm 1 (hard limit)	Support set audit
RL (PPO/DQN)	UNDECLARED 1 (reward)	Thm 1	$M > 1$ reward models
Imitation Learning	UNDECLARED 1 (expert)	Thm 1	Multi-expert audit
Recommendation	UNDECLARED ≈ 1	Thm 2	External feedback

表 3: Quick Reference: SCX Audit Summary. M_{eff} = effective multiplicity. "Violates" = which theorem/assumption is structurally violated. "Missing SCX" = the primary SCX component that would fix the deficiency.

Interpretation: Algorithms with $M_{\text{eff}} = 1$ are structurally unverifiable. No amount of benchmark accuracy can fix this. Algorithms with $M_{\text{eff}} > 1$ but undeclared have audit structure but lack formal certification. Only algorithms with declared M_{eff} and explicit Cercis Scores are SCX-compliant. Today, **zero** algorithms meet this standard.

参考文献

[1] SCX. *The SCX Framework: Structured Causal eXamination for Quality-Certified Learning*. 2026.

[2] SCX. *The SCX Audit Mandate: Why M-Parameter Declaration Must Be a Prerequisite for Scientific Publication*. 2026.

[3] SCX. *SCX Personal Ethics: Game-Theoretic Audit of Individual Honesty*. 2026.

[4] SCX. *SCX Audit of Governance: Multi-Expert Verification of Government Statistics*. 2026.

[5] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[6] L. Breiman. Bagging predictors. *Machine Learning*, 24(2):123–140, 1996.

[7] L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[8] Y. Freund & R. Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1):119–139, 1997.

[9] J. Friedman. Greedy function approximation: a gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

[10] T. Chen & C. Guestrin. XGBoost: A scalable tree boosting system. *KDD*, 2016.

[11] G. Ke et al. LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS*, 2017.

[12] D. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.

[13] A. Krizhevsky, I. Sutskever, G. Hinton. ImageNet classification with deep convolutional neural networks. *NeurIPS*, 2012.

[14] S. Hochreiter & J. Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

[15] K. He, X. Zhang, S. Ren, J. Sun. Deep residual learning for image recognition. *CVPR*, 2016.

[16] S. Ioffe & C. Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. *ICML*, 2015.

[17] N. Srivastava et al. Dropout: A simple way to prevent neural networks from overfitting. *JMLR*, 15(1):1929–1958, 2014.

[18] A. Vaswani et al. Attention is all you need. *NeurIPS*, 2017.

[19] J. Devlin et al. BERT: Pre-training of deep bidirectional transformers. *NAACL*, 2019.

[20] I. Goodfellow et al. Generative adversarial nets. *NeurIPS*, 2014.

[21] D. Kingma & M. Welling. Auto-encoding variational Bayes. *ICLR*, 2014.

[22] J. Ho, A. Jain, P. Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 2020.

[23] T. Chen et al. A simple framework for contrastive learning of visual representations. *ICML*, 2020.

[24] K. He et al. Momentum contrast for unsupervised visual representation learning. *CVPR*, 2020.

[25] K. He et al. Masked autoencoders are scalable vision learners. *CVPR*, 2022.

[26] C. Finn, P. Abbeel, S. Levine. Model-agnostic meta-learning for fast adaptation. *ICML*, 2017.

[27] V. Mnih et al. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015.

[28] J. Schulman et al. Proximal policy optimization algorithms. *arXiv:1707.06347*, 2017.

[29] T. Haarnoja et al. Soft actor-critic: Off-policy maximum entropy deep RL. *ICML*, 2018.

[30] S. Ross, G. Gordon, D. Bagnell. A reduction of imitation learning to no-regret online learning. *AISTATS*, 2011.

[31] D. Silver et al. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529:484–489, 2016.

[32] M. Spence. Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374, 1973.

[33] J. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005.

[34] Y. Gal & Z. Ghahramani. Dropout as a Bayesian approximation. *NeurIPS*, 2016.

[35] T. Miyato et al. Spectral normalization for generative adversarial networks. *ICLR*, 2018.

[36] P. Michel, O. Levy, G. Neubig. Are sixteen heads really better than one? *NeurIPS*, 2019.

[37] Y. Song et al. Score-based generative modeling through stochastic differential equations. *ICLR*, 2021.

[38] A. Radford et al. Learning transferable visual models from natural language supervision. *ICML*, 2021.

[39] T. Brown et al. Language models are few-shot learners. *NeurIPS*, 2020.

[40] J. Achiam et al. GPT-4 technical report. *arXiv:2303.08774*, 2023.

表 2: Complete Audit Verdicts —27 Algorithms, 27 Convictions

Algorithm (算法)	Verdict	§	Core Deficiency (核心缺陷)
Linear Regression (线性回归)	UNDECLARED	3.1	$M = 1$, single weight vector, no error detection
k-Nearest Neighbors (k 近邻)	UNDECLARED	3.2	Lazy state encoding, no SITUS Lipschitz bound
Naive Bayes (朴素贝叶斯)	UNDECLARED	3.3	Unverified independence assumption on features
Decision Tree (决策树)	UNDECLARED	3.4	$M = 1$, overfits silently, self-audit leaf probs
SVM (支持向量机)	UNDECLARED	3.5	Kernel SITUS without Lipschitz, no reconstruction
k-Means (K 均值)	UNDECLARED	3.6	Arbitrary K, self-referential WCSS metric
Bagging (装袋法)	PARTIAL	4.1	Bootstrap independence but no M_{eff} declaration
Random Forest (随机森林)	VERIFIED	4.2	Best classical; lacks explicit Cercis + Spring
XGBoost/Boosting (梯度提升)	UNDECLARED	4.3	Sequential experts violate A1, $M_{\text{eff}} \ll M$
Stacking (堆叠泛化)	PARTIAL	4.4	Meta-learner $M = 1$, unaudited weight learning
Deep MLP (深度多层感知机)	UNDECLARED	5.1	Depth without Theorem 3 audit, vanishing gradients
CNN (卷积网络)	PARTIAL	5.2	SITUS spatial encoding without Lipschitz guarantee
RNN/LSTM (循环网络)	UNDECLARED	5.3	Forget gate destroys evidence (Spring-1 violation)
ResNet (残差网络)	VERIFIED	5.4	Best deep arch; missing per-block verification
BatchNorm (批归一化)	VERIFIED	5.5	Satisfies A5; breaks at small batch sizes
Dropout	VERIFIED	5.6	2^n sub-networks voting; M_{eff} undeclared
Transformer (注意力)	VERIFIED	5.7	Multi-head = M queries; KV cache bounded memory
BERT (预训练语言模型)	UNDECLARED	5.8	Self-supervised $\to$ self-audit (Theorem 2)
GPT (自回归生成)	UNDECLARED	5.8	Autoregressive $M = 1$, hallucinations undetectable
GAN (生成对抗网络)	UNDECLARED	6.1	$M = 1$ discriminator $\to$ mode collapse, instability
VAE (变分自编码器)	UNDECLARED	6.2	Encoder-Decoder collusion = self-audit (Theorem 2)
Diffusion (扩散模型)	PARTIAL	6.3	Correlated denoising steps, $M_{\text{eff}} \approx 1$
SSL —Contrastive (对比自监督)	UNDECLARED	7.1	$M_{\text{aug}} > 1$ but unverified, augmentation destroys semantics
SSL —Reconstruction (重建自监督)	UNDECLARED	7.1	Pure self-audit, self-generated labels (Theorem 2)
Transfer Learning (迁移学习)	PARTIAL	7.2	No transfer quality metric, negative transfer undetected
Meta-Learning (元学习)	UNDECLARED	7.3	$M \leq K$ with class correlation, fundamental audit limit
RL / PPO (强化学习)	UNDECLARED	7.4	$M = 1$ reward, reward hacking, sample inefficiency
Imitation Learning (模仿学习)	UNDECLARED	7.5	$M = 1$ expert, distribution shift compounds silently
RecSys (推荐系统)	UNDECLARED	7.6	CTR feedback loop = self-audit (Theorem 2)